

Mapping & Value Chain Analysis of the Fruits and Vegetable Sub-Sectors in Somalia



TECHNICAL REPORT

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This technical report was conducted by UNIDO Somalia Programme team of experts, within the framework of the “Agro-technology development for Economic Growth in South and Central Somalia” project, implemented by UNIDO and funded by AICS/Italy.



Acronyms

1.	ODA	Overseas Development Assistance
2.	MSME	Micro Small & Medium Enterprises
3.	GDP	Gross Domestic Product
4.	FAO	Food & Agriculture Organization
5.	IMF	International Monetary Fund
6.	FSAU	Food Security Assessment Unit
7.	IDP	Internally Displaced Person
8.	IUCN	International Union for Conservation of Nature
9.	UNEP	United Nations Environment Programme
10.	MT	Metric Tonne
11.	USAID	United States Agency for International Development
12.	COOPI	Cooperazione Internazionale
13.	CEFA	European Committee for Training and Agriculture
14.	NGO	Non Governmental Organization
15.	OEC	Observatory of Economic Complexity
16.	EU	European Union
17.	FSNAU	Food Security & Nutrition Analysis Unit
18.	SATG	Somali Agricultural Technical Group

Contents

1. Background	5
1.1 Project Background	5
1.2 Socio-economic profile of Somalia	5
1.3 Scope of the Study.....	6
1.4 Methodology/ Approach of the study	7
a. Scoping visit to the field	7
b. Secondary Literature Review.....	7
c. Study Tools Development.....	7
d. Field research	7
e. Analysis and report writing.....	8
1.5 Scope & Limitation of the study	8
2. OVERVIEW OF AGRICULTURE SECTOR IN SOMALIA.....	9
2.1 Introduction.....	9
2.2 Land use pattern.....	11
2.3 Production and Cropped area of key crops	13
2.4 Market and its channels	13
2.5 Infrastructure scenario	14
2.5.1 Irrigation infrastructure	14
2.5.2 Storage infrastructure	16
3. FRUITS & VEGETABLE SUB-SECTOR IN SOMALIA	17
3.1 Production map.....	17
3.2 Agricultural Practices.....	20
4.2.1 Pre-harvest.....	20
4.2.2 Harvesting	24
4.2.3 Post Harvesting	26
4.2.4 Issues raised on the existing agriculture scene or need perception	29
3.3 Support Services	30
3.4 Processing or productive activities.....	31
3.5 Trading Channel & Sales	33
3.6 Sub-sector map.....	37
.....	37
4. VALUE CHAIN FRAMEWORK FOR FRUITS & VEGETABLES SUB-SECTOR.....	38
5.1 Governance.....	43

5.2 Inter-firm cooperation	44
5.2.1 Horizontal linkages.....	44
5.2.2 Vertical linkages	44
5.3 Structure of labour market.....	44
6 FINDINGS & KEY ISSUES	45
7 RECOMMENDATION ON AREAS for SKILL SET UPGRADATION	46
Bibliography.....	47

1. Background

1.1 Project Background

The Agro-Tech project which is being implemented by UNIDO is providing assistance in the revival of productive sectors in Central and South Somalia regions by establishing a network of three agro-technology and enterprise development units to support Somali businesses in starting, rehabilitating and upgrading agro-industrial operations, particularly by accessing to new technologies, markets, market-based skilled labour and financing facilities.

Emphasis in the project implementation will also be provided in supporting the labour force responding to the needs of the market with market based vocational and technical skills trainings to feed the growing needs of the MSMES sector with skilled labour capable of making modern and better tools, and in installing, operating, maintaining and repairing old and modern agriculture and agro-industrial equipment.

Furthermore, a credit facility to facilitate access to finance to project supported entrepreneurs is to be established and deployed within the framework of the project.

Overall, the project will aim at increasing the potential for economic opportunities and job creation in the agro-industrial sector of the Southern and Central regions of Somalia by focusing on four key actions:

- i. Promoting agro-technology upgrading and entrepreneurship development;
- ii. Delivering vocational and technical skill trainings to improve productivity of agro-industrial tradesmen and micro-industries
- iii. Facilitating access to technology, markets and finance; and
- iv. Providing institutional support in the area of agro-technology transfer, investment promotion, and entrepreneurship development.

In order to design the vocational training curriculum, the modules will be made based on skill gap analysis in three of the intervention sub-sectors- Fruits & Vegetables, Livestock and Fishery. Hence, to have an understanding of these three sub-sectors, a sub-sector mapping and value chain analysis is planned for each of these three sub-sectors and provide recommendation in the areas of the value chains where the skill gap analysis could look at. This project has been funded by the Italian Cooperazione with FGS- Ministry of Commerce & Industry (MoCI) as the line Ministry.

1.2 Socio-economic profile of Somalia

Facing decades' civil war, Somalia has gradually stabilized its economy with institutions and systems being built. The country is still fragile with highly dependent on aid and remittances, which is reflected when drought or major economic upheavals occur. As per the World Bank report the GDP of the country is growing at a steady rate (2 to 3 %) since 2013 with an exception of 2017 when due to a slump, the GDP shrunk. The 2018, nominal GDP for Somalia is **4.72 billion USD** with per capita GDP is 332 USD, which is not even 1 USD a day. The GDP estimate in Somalia is calculated using the expenditure method.

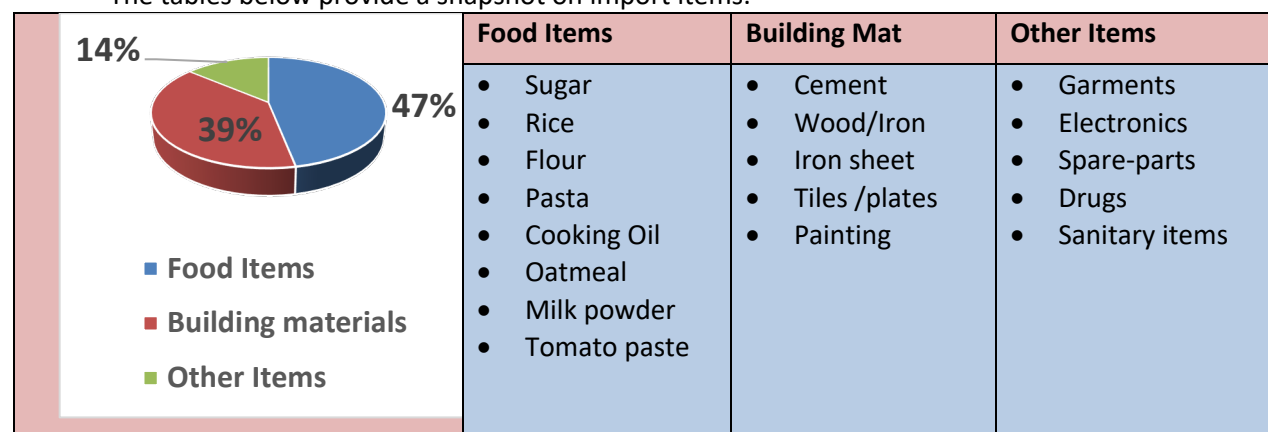
Item	Amount
Household final consumption expenditure	7,358
Government final consumption expenditure	531
Gross fixed capital formation	309
• Private investment spending	
• Public investment spending	
Net exports of goods and services	-3,478
	Export: 1,224
	Import: 4,702

All figures in million USD

It can be seen that the substantial contribution to GDP is house-hold consumption and not government expenditure or industrial production. The negative net exports shown above shows that the Imports is almost the same of GDP, which **indicates that the economy is highly import driven and is the main source for fuelling the house-hold consumption.**

An analysis of the import/export figures received from Mogadishu Port Authority (Fig. for year 2019).

The tables below provide a snapshot on import items.



It can be seen from the above table that food import constitutes a significant amount which is further substantiated by FAO-World Bank report that states Somalia faces chronic food crop deficit with own production can only 22% of the per capita needs.

The limited exports that goes from Somalia is highly agriculture oriented with 93% share is from Agriculture of which Livestock (Whole animal) presents 50% of the total. The other agriculture items are Sesame and dry lemon which is exported. Most of the live animal is exported to Saudi Arabia- specially catering to Hajj visitors and during Eid Al-Adha.

While, official remittances from the diaspora are estimated at about \$1.4 billion a year, equivalent to about 23 percent of Somalia's GDP (IMF 2017) and could be the major contributor to house-hold private consumption that is the back-bone of Somalia's GDP. About 3.4 million people—roughly 40 percent of the population—depend on remittances for their daily needs, and about 80 percent of all new business ventures are funded by remittances. The IMF figures for 2017 quotes that Somalia received foreign assistance as ODA is 1.65 billion USD.

Somalia, being import depended country with minimal exports, the balance of payment is highly skewed with trade deficit as mentioned in 2018 as 3478 million USD. This negative balance of payment is normally covered through foreign remittances (almost 25% of GDP) and external grants classified as ODA (Overseas Development Assistance).

1.3 Scope of the Study

- Mapping of the existing Fruits and Vegetable sub-sector in Somalia.
- Analyse various value chains operating within the sub-sector.
 - Production system
 - Demand and Supply Analysis
 - Value addition within the Chains
 - Marketing Channels
 - Actors/Sub-actors/Stake-holders
 - Horizontal and Vertical Linkages/Relationship among the Actors
 - Governance
 - Inter-firm cooperation

- Recommendations in selected areas of the operating value chains on the probable areas of skill set building.

1.4 Methodology/ Approach of the study

The methodology adopted was with twin pronged approach for conducting the studying which includes primary information collection & analysis and secondary data analysis for achieving the overall objectives.

The assessment of the fruits and vegetable subsector and its value chains was conducted by UNIDO, using the following steps:

- Scoping visit to the field
- Secondary literature review
- Study tools development
- Field research
- Analysis and report preparation

Each of these activities is detailed below:

a. Scoping visit to the field

The first step taken by the research team was to acquaint themselves with the ground realities and understand the broad parameters of the sub-sector operating in the country. The team interacted with various experts, key informants of the sub-sector and prepared a preliminary report, which was shared with the PMU.

This scoping visit helped the research team to have a preliminary idea about the various players in the value chain. This was useful in designing the various questionnaires that were used to collect data from field.

b. Secondary Literature Review

Different secondary literature was collected and reviewed to develop a general idea on the sub-sector and its associated value chains and the study area. Extensive web search was done to locate secondary information that would be useful for analysis. The secondary literature was continuously referred throughout the primary data collection as well as during analysis and report writing period. In addition, inputs from expert opinions from various institutions have been collected and used in the study.

A list of prominent articles and documents that were referred to has been provided as part of the annex

c. Study Tools Development

As a part of the primary data collection, it was decided to have individual survey of the value chain actors. **The individual survey was conducted with fruits & vegetable farmers, traders, processors.** Separate sets of questionnaire/ checklist were developed for each actor and used in the field research.

d. Field research

The field research component was conducted for a period of 30 days. During this period, individual interview with fruits and vegetable producers, individual interviews with traders, individual interviews with key informants including processors of fruits and vegetables were conducted by team consisting of local expert and UNIDO local staff.

The interaction with stakeholders focused upon tapping the knowledge base of the local stakeholders, the status of fruits and vegetable production in the country and also assesses their capacities for involvement in the activity.

The interactions and visits were supported by analysis of the secondary data related to fruits and vegetable production and trade, and also helped in forecast and estimation.

e. Analysis and report writing

All the data collected from individual interviews were then fed into MS Excel sheets from the hard copies of the questionnaire. These data were then used to generate tables, which were analyzed to find the situation and the status of the value chain. These tables are then incorporated into the report, and the combination of the data analyses of primary data and inferences from the secondary literature reviewed coupled with opinion from key informants were the basis for the report.

1.5 Scope & Limitation of the study

Some of the limitations faced by the study team during its course of work is reported below:

- The sample size for primary data has been selected mostly from Banadir and Juba-land, where accessibility of the research team was high. The input received from these samples along with secondary data has been extrapolated to generalise for the country, which might have overlooked many local conditions prevalent in other regions.
- The primary data collection has been through re-call methodology, which has the limitation due to non-documentation of records as well as the fear of mixing of records pertaining to other crops.
- Fruits and Vegetable cultivation and trading is a highly un-organized economic activity, hence the authenticity of secondary data used is at times doubtful.
- Somalia has come out of civil war only few years back. Many of the research Institutions as well as Government entities for data collection are still in their infancy. Due to this situation, the availability of secondary data was very difficult and limited. Even, whatever secondary data reported, the authenticity is not beyond doubt.

2. OVERVIEW OF AGRICULTURE SECTOR IN SOMALIA

2.1 Introduction

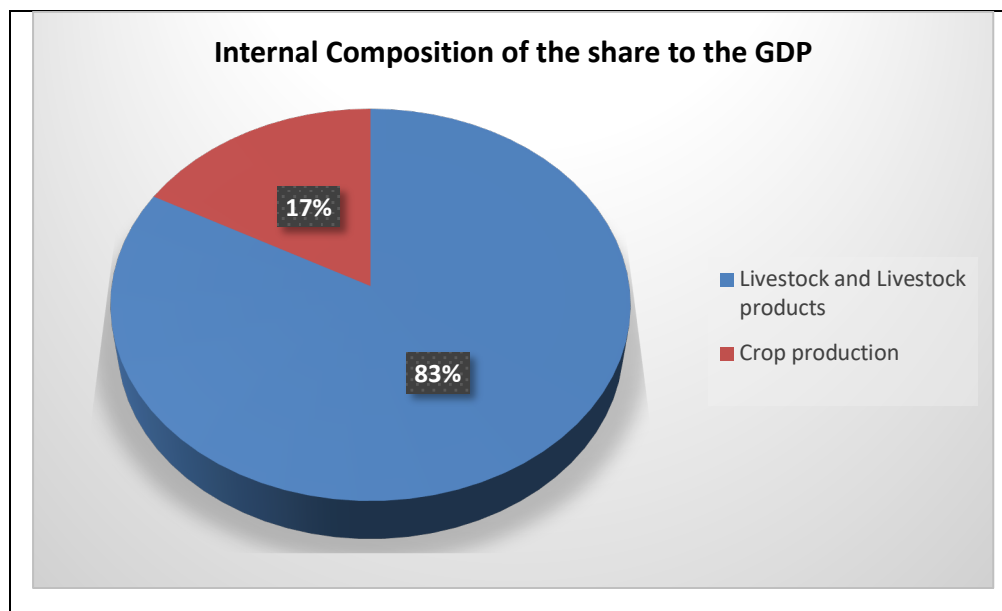
The sub-sector mapping and value chain report before analysing and reporting on the fruits and vegetable sub-sector provides an understanding of the whole Agriculture sector to which Fruits and Vegetables are part of and its associate support structures.

Primarily when one delves into the Agriculture sector, it needs to be understood that there are various components included in it. As far the classification in terms of Somalia are the following:

- **Livestock and Livestock production**
 - Meat
 - Whole animal
 - Raw milk
 - Raw hides
- **Crop production**
 - Staple food crop
 - Fruits
 - Vegetables
 - Forestry
 - Others
- **Fishery (But negligible in Somalia)**

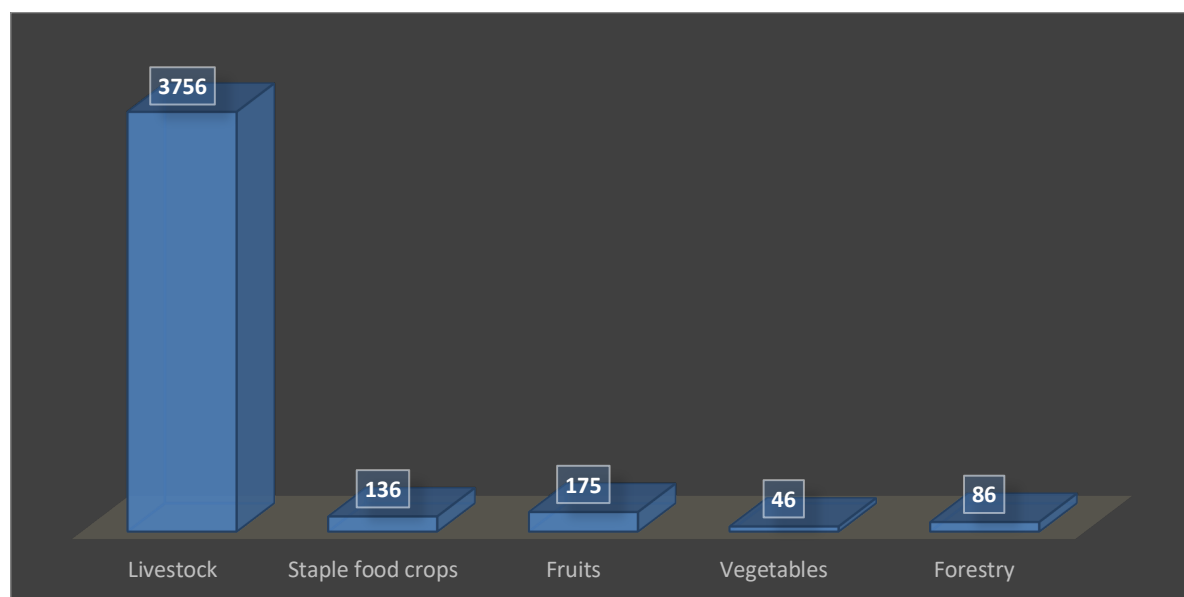
Roughly speaking, the share of the components in the GDP of the country is reflected below:

Figure 1: Composition of Agriculture sub-components



Source: Somalia- Country Economic Memorandum – Vol I , FAO and World Bank

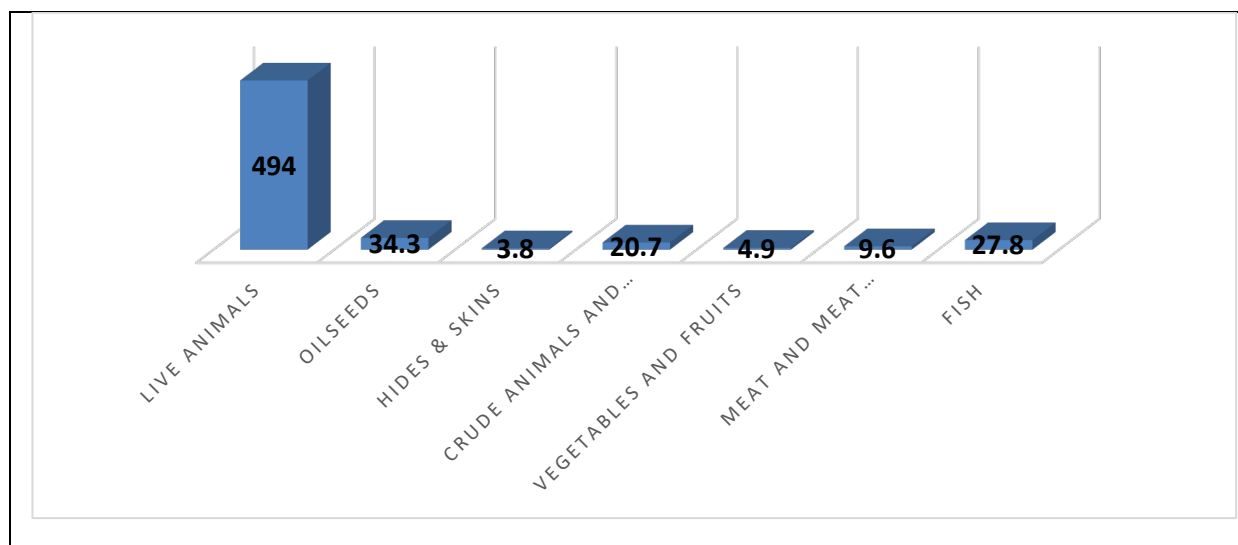
Figure 2: Gross production value of livestock & crops in Somalia-2013-16 (Million US\$)



Source: Somalia- Country Economic Memorandum – Vol I, FAO and World Bank

Agricultural exports are prominent component in the overall country's total export and is valued to be \$634 million, which is more than five times the value before the civil war, thanks to livestock exports, which rose by a factor of almost 10 since the late 1980s.

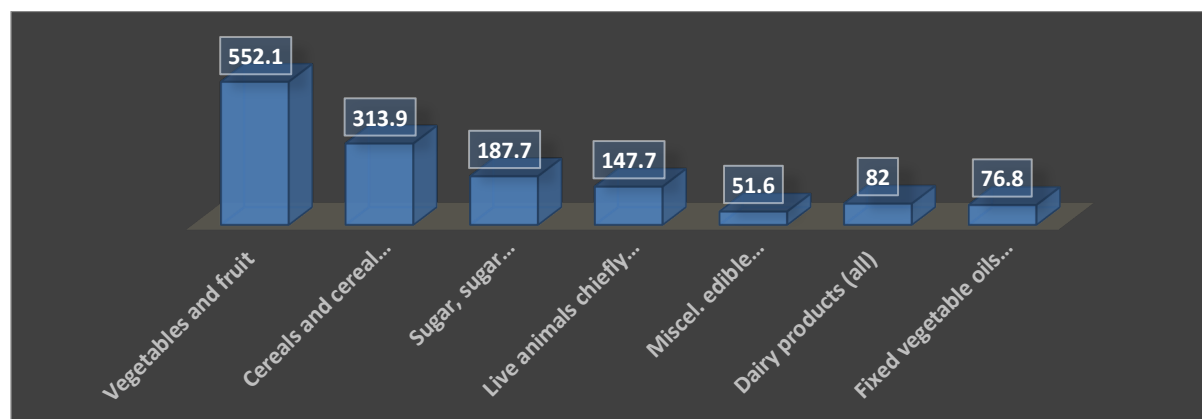
Figure 3: Top export items in the Agriculture sector for the year 2015 (In million US\$)



Source: Somalia- Country Economic Memorandum – Vol I, FAO and World Bank

Similarly, Agricultural imports have also risen and have reached almost \$1.5 billion in 2015. A break-up on the different categories of agro- imports for the year 2015 is presented below:

Figure 4: Top import items in the Agriculture sector for the year 2015 (In million US\$)



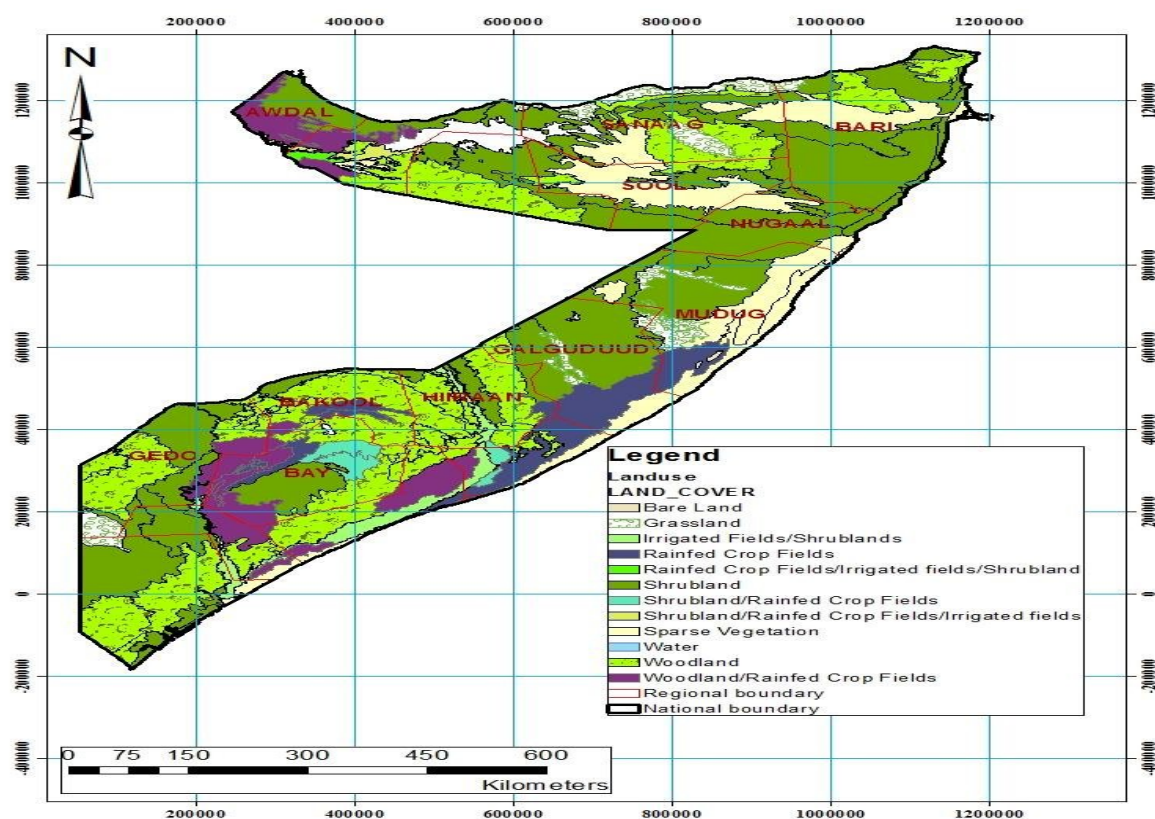
Source: Somalia- Country Economic Memorandum – Vol I, FAO and World Bank

Only 5% of the total land area Somalia (3 million hectare) is cultivable, out of which only 0.7 million ha is irrigated. With growing population, increased internal displacement has led to chronic food crop deficit. According to FAO and World Bank group report, local crop production meets only 22 % of the total requirement, which may go up to 40-50 percent of per capita cereal needs in best of the Agricultural seasons.

2.2 Land use pattern

The map below provides a snapshot of the land use pattern of Somalia:

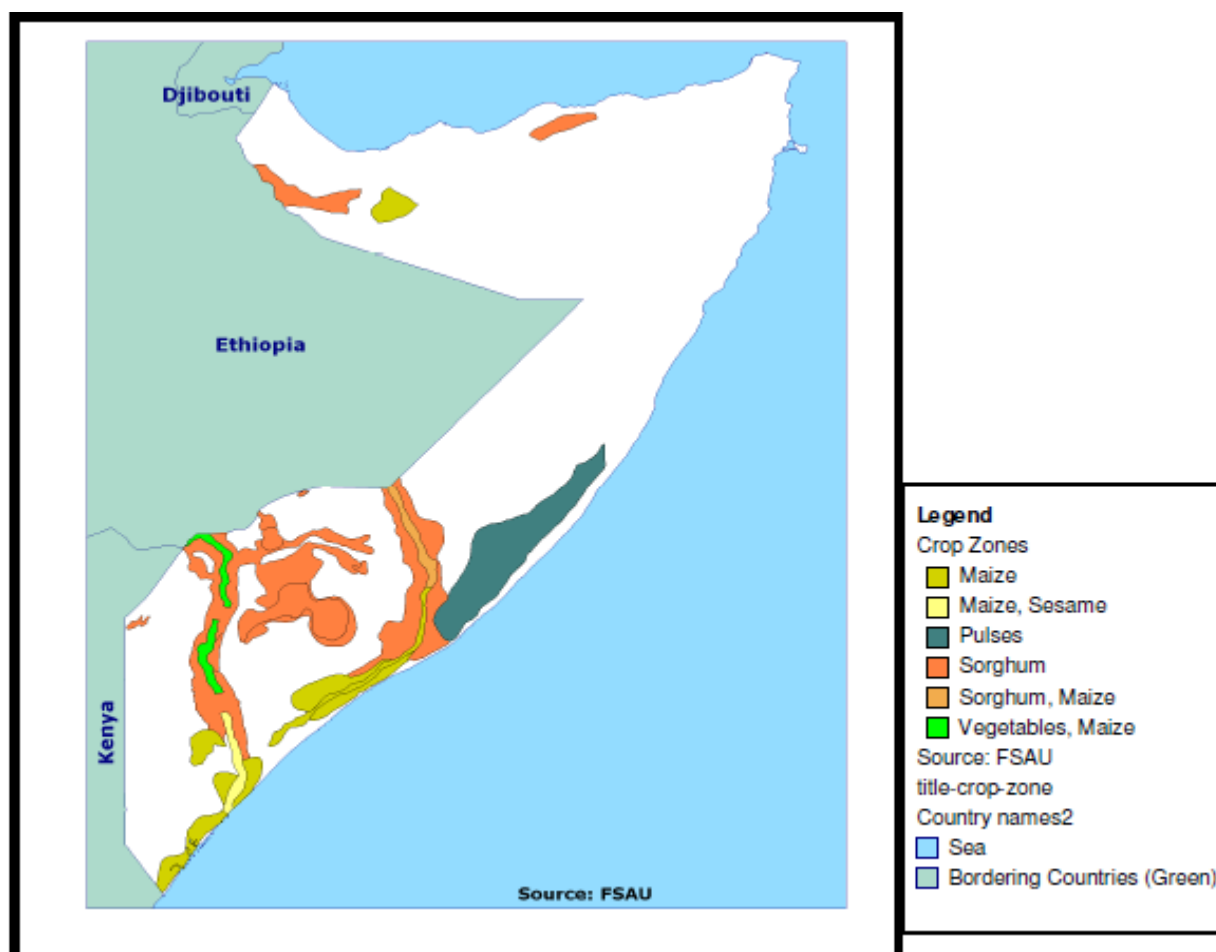
Figure 5: Land use pattern of Somalia



According to research article on Journal of Agricultural Informatics written by **Mark K. Boitt , Flomena C. Langat , John K. Kapoi** - Somalia's total land area is 637,540 km² , of which 30% is classified as desert land unsuitable for agricultural production, 45% is covered by rangelands suitable for livestock grazing, 14% is covered by forest or woodland, and the remaining 11% is classified as arable land.

Most crop production is undertaken in the Southern regions by small-scale subsistence farmers with an average of 0.2–3.0 hectares of land. According to the FAO, the smallholder farming subsector accounts for 80 percent of total crop output and 70 percent of marketed agricultural produce. The cropping zones in the country is presented below:

Figure 6: Crop zones of Somalia



Source: Food Security Assessment Unit (FSAU)-Somalia

The map clearly reflects that most of the cropping zones are in the south of the country, which forms the region through which the rivers of Juba and Shabelle passes through.

There are four primary agricultural zones in Somalia:

- Northwest in parts of Awdal and W. Galbeed – rain-fed maize and sorghum.
- Coastal Cowpea Belt Zone in Central and Southern **Somalia**.

- Shabelle and Juba Riverine Valleys – rain-fed and irrigated maize, with sesame cash crops.
- Sorghum Belt in Bay and Bakool Region – rain-fed sorghum with livestock production.

2.3 Production and Cropped area of key crops

The study using secondary data source has compiled the production data of the major cereal crops, oilseeds, vegetables and fruits.

Table 1: Agricultural crops production figure 2018

Name of the crop	Production Volume (MT)	Cropped Area (ha)	Yield (hectogram/ha)
Maize	138716	92920	14929
Sesame seed	25788	46489	5547
Sorghum	129523	234403	5526
Sugar cane	209291	5537	378005
Bananas	1226	20905	170478
Beans, dry	26855	87892	3055
Cassava	92875	9695	95794
Coconuts	11074	3962	27951
Dates	13785	2666	51715
Fruit, fresh	143063	12975	110258
Grapefruit (incl. pomelos)	5442	1105	49255
Lemons and limes	7960	1312	60682
Tomatoes	23962	16276	14722
Vegetables, fresh	75382	6994	107775
Watermelons	6105	633	96367

Source: FAOSTAT

2.4 Market and its channels

The main market actors in the agriculture trading in Somalia consists of importers, wholesalers, local trader's shopkeepers and open-air retailers. As mentioned earlier in the report, the majority of the cereal crops is imported, where the major players are the importers who brings it in mostly through Mogadishu, Berbera and Bosaso ports. The imported items are then stored in mostly agro-warehouses owned by private players. The wholesalers procure from these importers and takes it forward for sales. Some of these importers are also engaged in wholesale and retail activities to increase the turnover. These importers are also contacted by humanitarian agencies like WFP for direct transfer of food grains to beneficiaries.

There are three types of transactional arrangements: Importers imports and sells their own network of wholesalers and retailers or sell to selected wholesalers in Somalia based on prior arrangements or sell directly to traders/retailers. As for the wholesalers, some of them handle imported food items procured from importers and sell to retailers and others are those dealing locally-produced food items procured from local traders/farmers.

Retailers/Shopkeepers engaged in the retail sales. They operate from fixed locations (either permanent or temporary) in specific main or satellite markets. They procure their supplies from wholesalers operating in the same markets or pool resources to buy supplies (sugar and rice) from importers at the port.

There is another category of market player - Open air retailers, predominantly women, who trade in open spaces or under temporary structures. They move from one area to the other within the same district. Many of them are IDPs. During harvest period, they procure from farmers while in lean season, they procure from wholesalers/ local traders.

Transfer of commodities between market players primarily takes place in two places, the ware-houses in the ports from wholesalers buys from importers and secondly in designated regional markets, where wholesalers display their items to be purchase by local traders/ retailers.

Table 2: List of regional markets in Somalia

Regions	Markets	Regions	Markets
Bay	Baidoa	Lower Shabelle	Merca
	Diinsor		Qorioley
	Qansax-dheere		Afgooye
Bakool	Xudur		Wanle-Wayne
	ElBarde	Middle Shabelle	Adan-Yabal
Hiraan	Belet-Weyne		Jowhar
Gedo	Bardera	Middle Juba	Buaale
	Doolow		Jilib
	Belet-Xaawa		Afmadow
	El Wak	Lower Juba	Jamame
	Luuq		Kismayo
Banadir	Mogadishu	Bari	Bossaso
Awdal	Zeltac	Wog. Galbeed	Hargeisa
	Borama	Nugal	Garowe

2.5 Infrastructure scenario

2.5.1 Irrigation infrastructure

Roughly, 25% of the crop production is through irrigation while rest 75% is through rain-fed farming. It is prevalent across the country with exception in the coastal sandy plains and high limestone areas. The rain-fed farming is done in two rainy seasons, **gu, or main rainy season**, lasts from April to June; the **second rainy season, called the dayr**, extends from October to December. The coastal region in the south around Mogadishu and Kismaayo has an additional rainy season, the **Xagaaye** (July to August), in which isolated rain showers occur. The areas of the Juba and Shabelle basins are 218 114 km² (to Jamaame, excluding Shabelle basin) and 296 972 km² (to the Juba confluence). The major rain fed crops grown in Somalia are Sorghum, Cow pea and some limited maize.

Irrigated agriculture is practiced in the floodplains along the permanent rivers in south Somalia (the Juba and the Shabelle) and along the seasonal streams and springs. In northern Somalia, water is available within pockets of deep soil for irrigated orchards, or from shallow wells and springs, which are the major sources of water for crop irrigation, with water pumped to the fields. **Irrigated farming systems are based mainly on small and medium-scale gravity and pump irrigation.** Irrigated crops grown on a small scale include maize, sesame, fruit trees and vegetables, while crops such as bananas, guava, lemon, mango and papaya are grown on a large-scale for domestic consumption.

Figure 7: Irrigation schemes & infrastructure along Juba river

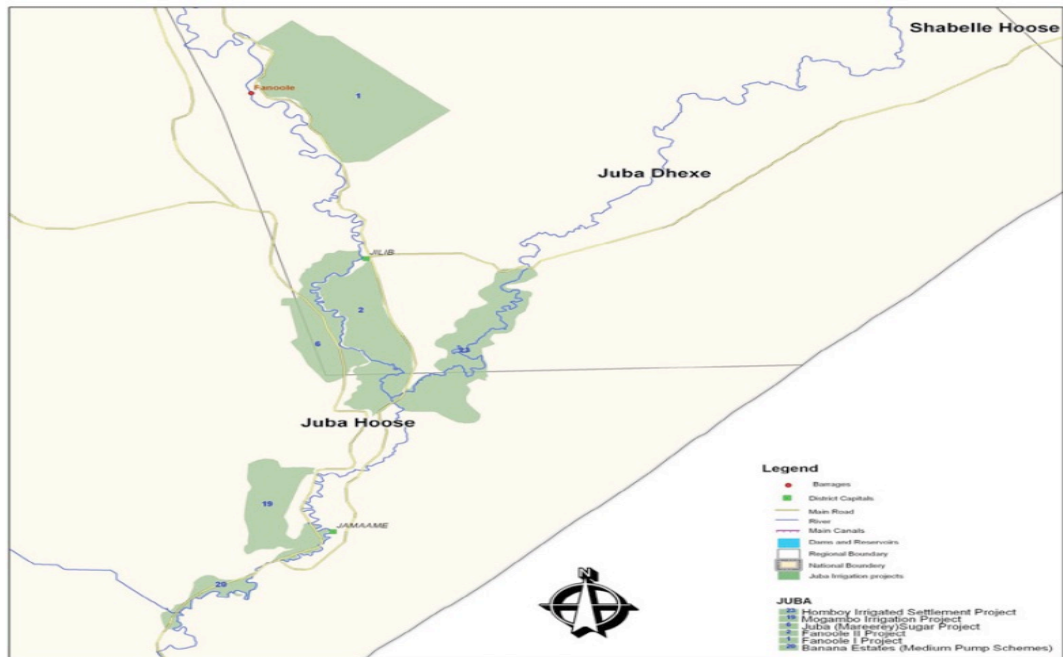
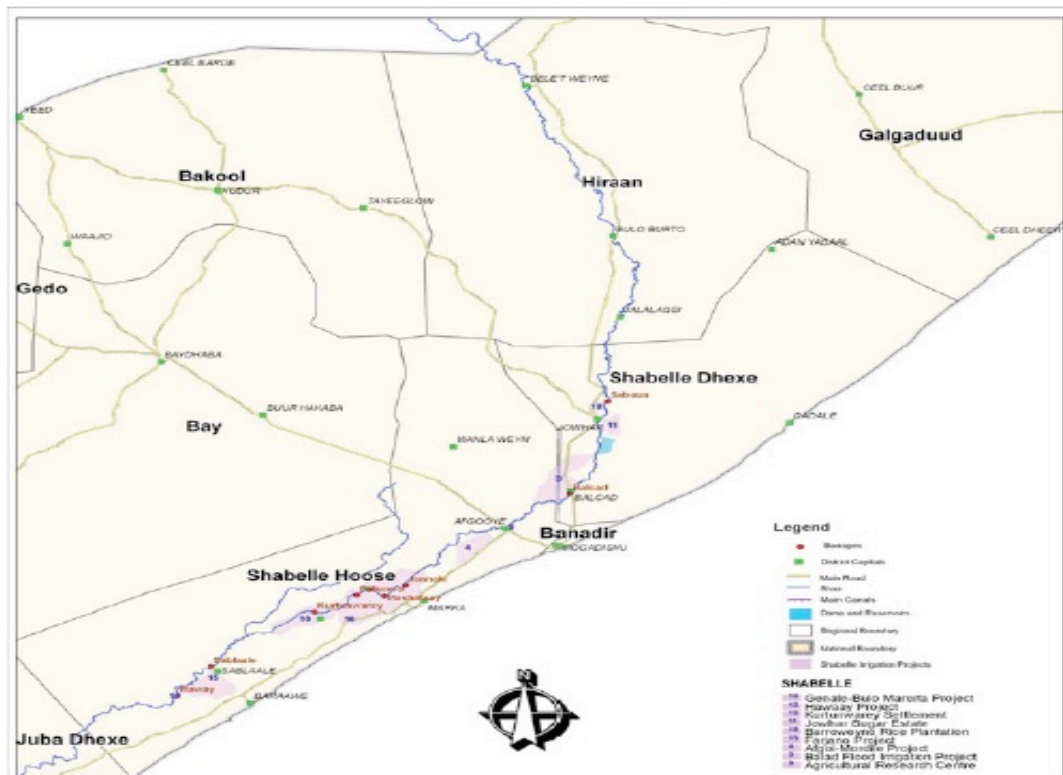


Figure 8: Irrigation schemes & infrastructure along Shebelle river



Oasis Farming is practiced in Somaliland and Puntland and traditionally occurs in **toggas** and springs which provide water for the production of date palms, papaya, citrus and vegetables.

2.5.2 Storage infrastructure

Agricultural production in Somalia suffers greatly from post-harvest and storage grain losses. Almost 20-30% of the total harvest is lost due to storage losses which translates to the order of 50,000 to 80,000 tonnes per year. This loss is mainly due to the post harvesting practices followed by farmers in Somalia. A survey conducted by SATG in the Bay region reveals that grain losses associated with the traditional storage systems (underground pits) are significantly higher than those associated with post-harvest techniques (harvesting, transportation, and drying), at an estimated 40% and 20% respectively. In order to reduce storage losses, SATG is promoting the design of a locally popular metal water-storage tank to create a metal grain silo which can easily be manufactured within the target communities.

As for the ware-housing facilities for the traders and importers, most of the facilities are owned by private operators or humanitarian agencies. A snapshot of all the ware-housing facilities available in Somalia is presented below:

Table 3: Mapping of the ware-housing facilities in Somalia

Type	Nos.	Capacity	Location
Govt. owned/ Agency owned	3	15000 MT each	Mogadishu Port
Commercial- Private owned	4	NA	Mogadishu
Humanitarian- Agency owned	NA	8640 sq. m.	Galkayo
Humanitarian- Agency owned	NA	4800 sq. m.	Dolow
Humanitarian- Agency owned	NA	400 sq. m.	El Wak
Humanitarian- Agency owned	NA	400 sq. m.	Dobley
Humanitarian- Agency owned	NA	800 sq. m.	Kisimayo
Humanitarian- Agency owned	2	300 sq. m. & 75 sq. m.	Baidoa
Humanitarian- Agency owned	2	150 sq. m. each	Jowhar
Humanitarian- Agency owned	NA	31000 MT	Hargeisa and Berbera
Humanitarian- Agency owned	NA	5000 MT	Hargeisa
Humanitarian- Agency owned	NA	680 MT	Hargeisa
Commercial- Private owned	2	12000 MT	Hargeisa
Commercial- Private owned	NA	99988 MT	Hargeisa
Commercial- Private owned	4	51000 MT	Hargeisa
Commercial- Private owned	NA	149000 MT	Hargeisa

Source: Somalia NGO Consortium

3. FRUITS & VEGETABLE SUB-SECTOR IN SOMALIA

As discussed earlier, the fruits and vegetables sub-sectors contribution to farming activities is significant and needs an analysis and understanding to intervene so that its contribution can further grow and become economically a significant component in the GDP of the country.

3.1 Production map

A mapping of the various fruits and vegetables grown in the country and the respective regions were made based on review of secondary data and inputs received after series discussions made with various agricultural experts in the country.

Table 4: Map of the geographical areas of various fruits and vegetables

Name of the Fruits & Vegetables, and Oil-seeds	Approx. nos. of growers in the country¹	Prominent geographical areas where these are grown
Dates	2000	Lower and Middle Shabelle as well as Juba regions
Water melon	200,000	Juba region, Shabelle, Bay and Bakol, Galgadud, Puntland and Somaliland
Beans (nut)	10,000	Middle Shabelle, Lower, Galgadud, Puntland, Somaliland, and Juba and Bay and Bakol
Coconut (Can be fruit, nut and seed)	20,000	Middle Shabelle, Lower, Galgadud, Puntland, Somaliland, and Juba and Bay and Bakol
Pepper	2000	The shabelles and Juba regions
Peanut (nuts)	100,000	Lower shabelle, middle and some part of Juba region and the oasis of part of Puntland
Lemon	325000	The lower and middle Shabelles, Juba region, Part of Somaliland, Puntland, Bay and Bakol, Galgadud
Tomatoes	200,000	The lower and middle Shabelles, Juba region, Part of Somaliland, Puntland, Bay and Bakol, Galgadud
Mango	200,000	Lower shabelle and Middle shabelle and Part of Ged region in Jubaland state
Pineapple	200,000	Middle shabelle, Lower shabelle and Part of Juba region
Orange	6000	Lower shabelle, middle and some part of Juba region
Papaya	10,000	Middle shabelle, Lower shabelle and Part of Juba region
Banana	200,000	Majorly from the lower and middle Shabelles and Juba Regions
Peas (legume vegetables)	5000	Lower shabelle , Middle and part of Juba and the oasis area of Puntland
Carrots	10,000	The lower and middle shabelles and part of Jubaland's regions

¹ Numbers have been estimated after reviewing various secondary reports and also on discussion with Somalian agricultural experts.

Name of the Fruits & Vegetables, and Oil-seeds	Approx. nos. of growers in the country ¹	Prominent geographical areas where these are grown
Pumpkin	100,000	The lower and middle shabelles and part of Jubaland's regions
Spinach	80,000	Middle shabelle, lower shabelle, lower juba, Gedo part of Bay and Bakol and Somaliland
Sesame	200,000	The lower and middle Shabelles, Juba region, Part of Somaliland, Puntland, Bay and Bakol, Galgadud
Onion	5000	Middle shabelle, Lower shabelle, somaliland, Part of Puntland, Gagaduud, Bay and Bakol and Juba regions of Jubaland

Characteristics of the prominent fruits and vegetables grown in Somalia are reported here to have an understanding of the type and varieties grown.

Table 5: Characteristics of some fruits and vegetables in Somalia

Name of the fruit & vegetable	Characteristics
Grape-fruit	The grape fruit tree, grows best on well drained soils.
	Rainfall should amount to at least 1200mm / year, and dry periods should not exceed 2 months, otherwise additional irrigation becomes necessary, as is the case in Somalia.
	At the age of 6 years a grapefruit tree starts to bear fruits .The most productive period is reached at the age of 10 years and lasts for another 15 years.
	There are two harvest seasons , the first around April to June and the second around December to January.
	There exist 7 varieties in Somalia, six of them with white or pink coloured flesh and 1. The most prominent ones are: Shamba, Thompson and Red-plush
Lime	The commercial cultivation of lime trees started in Somalia in the mid- 1960s in the Afgoy area.
	Even though there are two peak seasons (June and December), lime trees yield more or less all the year round.
	The content of one sack of lime is around 1,800 pieces of lime and the average yield is around 2 sacks of 50 kg / plant / year.
Papaya	There exist two main varieties : the traditional tall trees, which had been introduced by the Italians and a dwarf variety which was introduced more recently, and which has gained ground to its convenient way of harvesting.
	The tree starts bearing fruits after 6 to 7 months (and 8 to 9 months for the tall variety) and will continue fruiting for about three years.
	In one year a Papaya tree produces around 2 boxes and the number of trees per hectare, is around 800.
Bananas	Originally the Jubba variety had prevailed by far, but it has been gradually replaced by the Poyo variety .

Name of the fruit & vegetable	Characteristics
	The average yield amounts to 300 qtls / ha / year.
Mango	The mango tree is a typical representative of the tropics. At temperatures below 19 degrees Celsius, growth and productivity decreases significantly.
	The minimum amount of rainfall has to be at least 1,000 mm per year; otherwise artificial irrigation becomes necessary.
	In Somalia where two rainy seasons occur: Gu (April – May) and Deyr (October – December) the tree bears fruits even twice a year.
	In Somalia alone, more than 10 varieties are cultivated. The most common ones are: Borbo, Dodo, Shushunle . At present the most popular variety is “Borbo”, followed by “Dodo”.
	On average a fully productive tree “Borbo” type yields about 1,000 fruits per year, while the “Dodo” variety produces 1,200 fruits per year.
Watermelon	The watermelon is a dry season crop in Shabelle regions. They are largely grown either in rain-fed or under residual soil moisture of the flooded land and to a lesser extent under regular irrigation.
	Charleton Grey, Crimson Sweet, Sugar Baby are the most common varieties grown in Somalia.
	About 1 kg of seed is sufficient for one hectare of land area. December to January is the most suitable period of watermelon sowing
	Harvesting starts 90 to 95 days after emergence. Around 10 to 12 tons/ Ha are harvested depending on crop management, cultivator and environmental factors.
Carrot	The crop is grown under irrigation at Afgoy and Marka districts.
	Different varieties in Somalia are Chantenay, Nantes coreless, Nantes
	The first week of May and all of July are regarded favourable periods for planting. Sowing can be extended beyond July up to October.
	Harvesting starts 80 days after emergence. It continues for up to one month depending on market demand. Carrots yield approximately 25 tons/ Ha
Tomatoes	Tomatoes are grown under irrigation; rain fed and water recession of the flooded land at the Shabelle and Juba regions.
	Local Cherry, Shalambot, Roma VF, San Marzano are the main varieties grown.
	The first week of March, marks the beginning of tomato seeding. Seedlings are transplanted 30 days after emergence. April and May are favourable periods for transplanting irrigated tomatoes.
	Harvesting starts 60 to 70 days after transplanting and continues up to two months with successive yield reductions. Yield of tomatoes varies from 0.5 to 9.0 tons / Ha depending on the crop management, varieties and environmental factors.
Peppers	Peppers are normally grown under irrigation and are readily available at any time of the year at local markets. They are classified as “sweet” and “hot”.

Name of the fruit & vegetable	Characteristics
	California Yellow Wander, Cayenne Long Slim are two common varieties grown.
	At early stages of crop establishment, weeding needs to be carried at intervals of 4 to 5 days till the crop attains its firm establishment. Harvesting starts 60 days after transplanting and continues up to one year.

Source and Text: FEWSNET Horticultural study in Lower and Middle Shabelle.

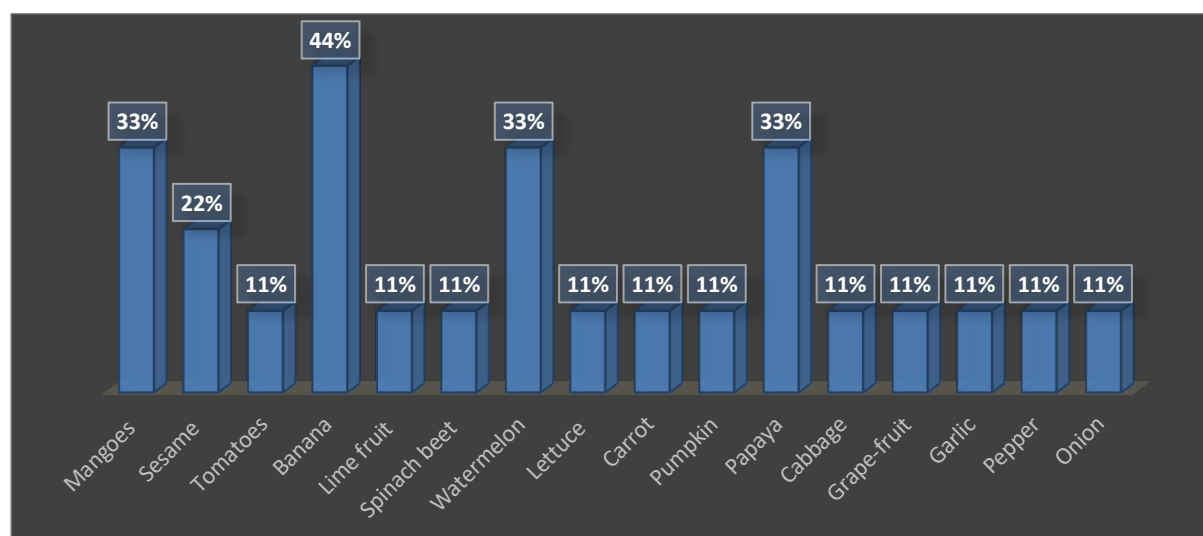
3.2 Agricultural Practices

The report tries to understand the existing agricultural practices of the farmers involved in fruits and vegetable cultivation in Somalia. It is classified into three categories – Pre-harvest, harvest and post-harvest.

4.2.1 Pre-harvest

- **Cropping pattern**

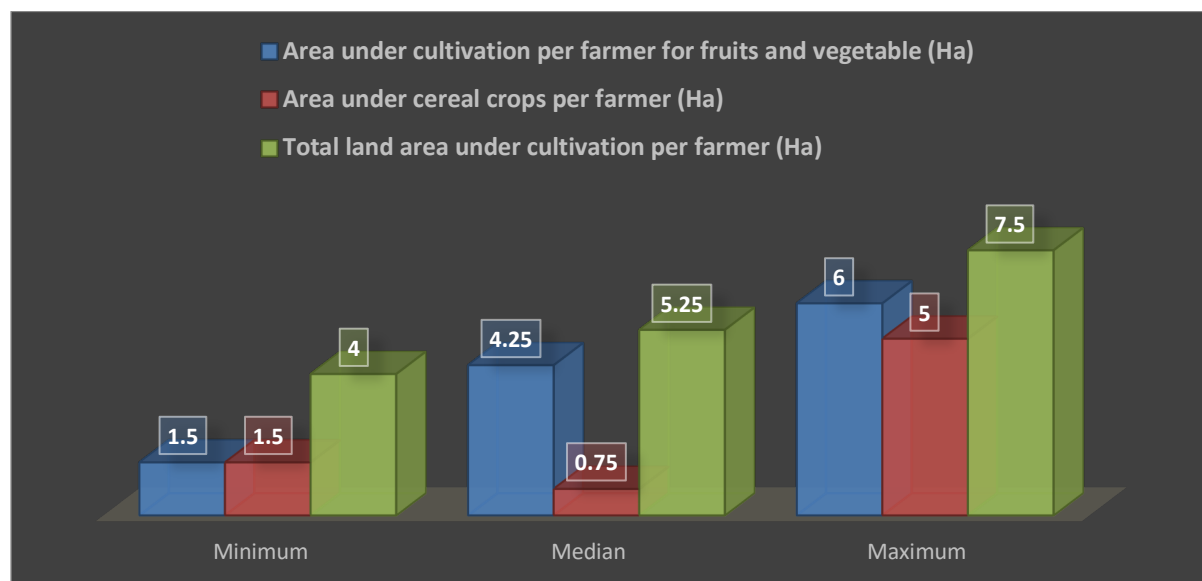
Figure 9: Cropping pattern followed by farmers



It can be seen that amongst the farmers interviewed that preference for growing fruits is high with Bananas, Papayas, Water-melon and Mangoes being the most preferred. Though not reported in the study, most of the respondents (50%) grows cereal crops like Maize and Sorghum. As seen in the figure above, though there is no preferred vegetable amongst the respondents but almost of them grows one or the other vegetables. Though it cannot be confirmed, maybe the preference for selected fruits could be due to assured market or higher margin. Maybe, the vegetables are locally sold with lower margins, the resource constraints of water may push for higher margin crops.

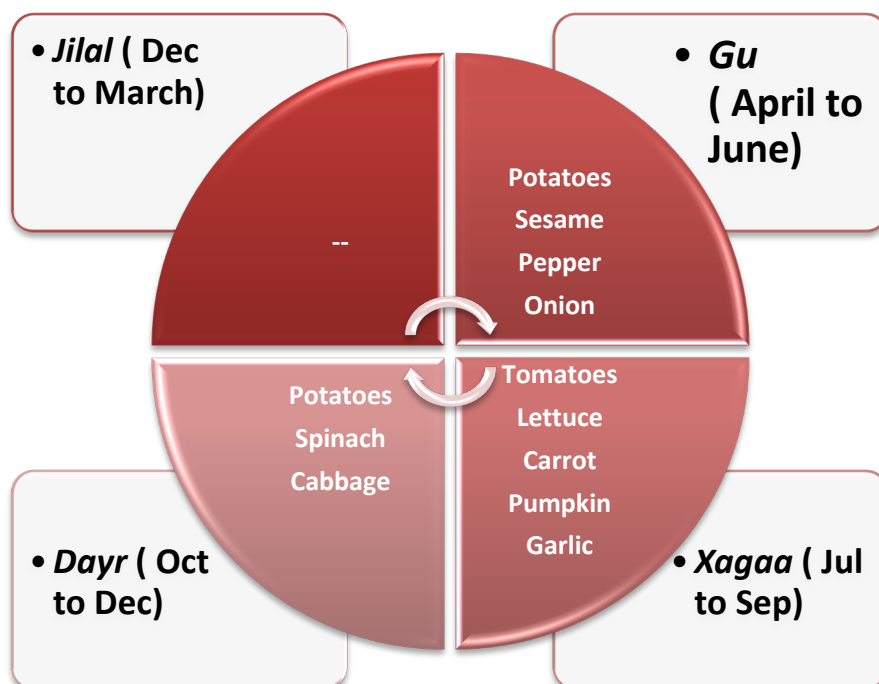
The average area of land utilized by the farmers for cultivation is represented below:

Figure 10: Area under cultivation



The figure below presents the seasonal pattern of vegetables grown as per the respondents.

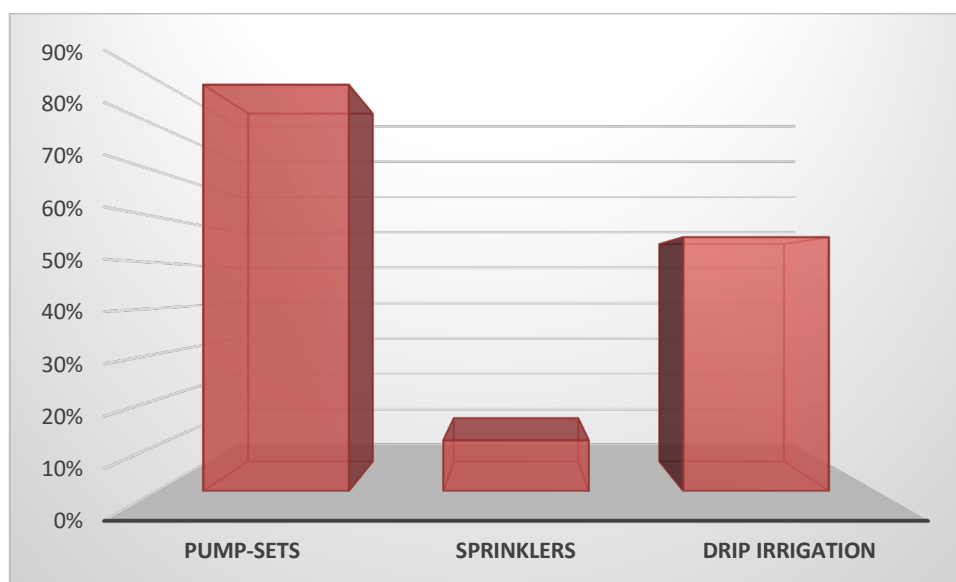
Figure 11: Seasonal cropping pattern of vegetables



- **Source of Irrigation**

All the respondents have mentioned that primarily the “Canals” are the source of irrigation. It could be that the respondents are all from the South Somalia in the river basins of Juba and Shabelle.

Figure 12: Tools and Equipment usage for irrigation



It can be seen that most of the farmers are using pump-sets to spread the water availability for the crops. Use of modern technologies like drip irrigation and in one or two cases of sprinklers indicates the progressive farming practices.

- **Volume of Production**

Fruits/Vegetables	Average land area under cultivation (ha)	Average production (kg)	Maximum production (kg)	Minimum production (kg)	Price per kg (US\$/kg)	Average income (US\$)
Mangoes	1.67	1500	1800	1200	1.5	2250
Sesame	2.5	2350	2700	2000	2	4700
Banana	1.125	1300	1500	1200	0.5	650
Lime	1	900	1000	800	0.6	540
Water-melon	1	1000	1500	800	1	1000
Papaya	1.67	1550	1200	1900	1	1500
Grape-fruit	1.5	1100	1200	1000	0.5	550
Tomatoes	1	1000	1000	1000	1	1000
Carrot	1	800	800	800	1.5	1200
Pumpkin	0.5	1000	1000	1000	1	1000

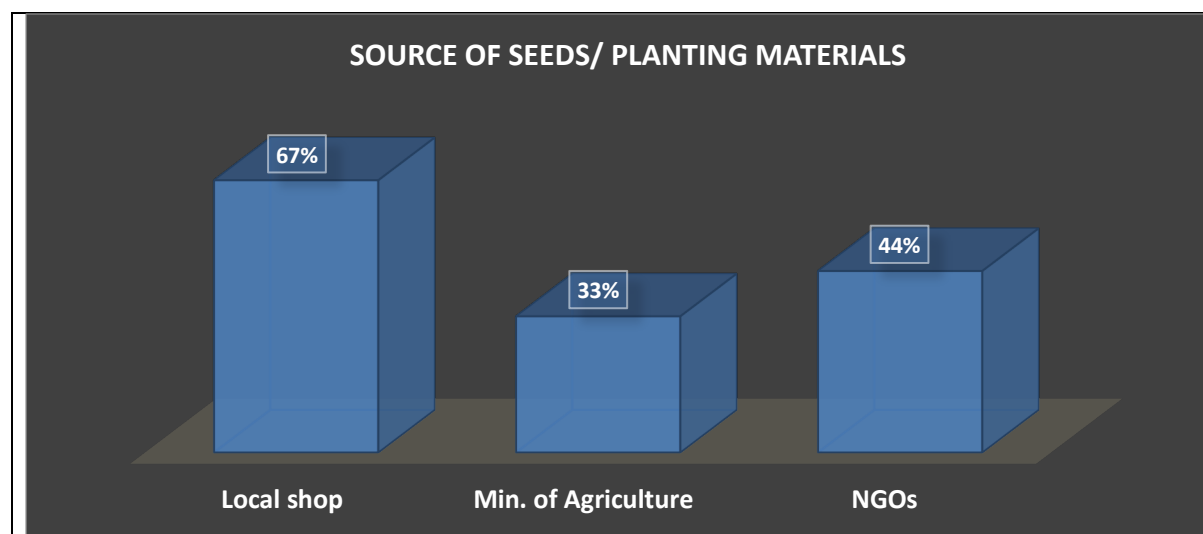
- **Management practices**

An overview of the existing agricultural management practices followed by the respondent farmers was analysed. The findings are reported below:

Figure 13: Management Practices - Part 1



Figure 14: Management Practices -Part II



Analysing the above two figures provides a fair perspective of the type of agriculture that is being pursued by the farmers interviewed. It is clear that farming practices are not obsolete and traditional. Farmers are using modern technology like tractors, spraying machines, best practices of using fertilisers/ pesticides, certified seeds etc. This indicates that productivity of the crops are important factors for a farmer and cultivation of fruits & vegetables are taken as commercial ventures.

Nevertheless, there is still scope of modernization for sustainable agriculture and increased production in view of very poor response towards soil testing for providing adequate and right mixture of fertilizers as well as knowledge on increased production using hybrid seeds.

4.2.2 Harvesting

A rough estimation is made on the probable months of harvesting of fruits and vegetables so that an assessment can be made on the period in which the farmers have cash flow for the sale proceeds:

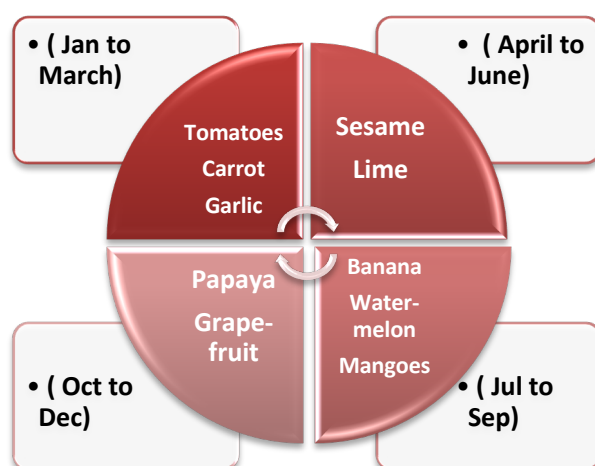


Figure 15: Plucking and harvesting methods

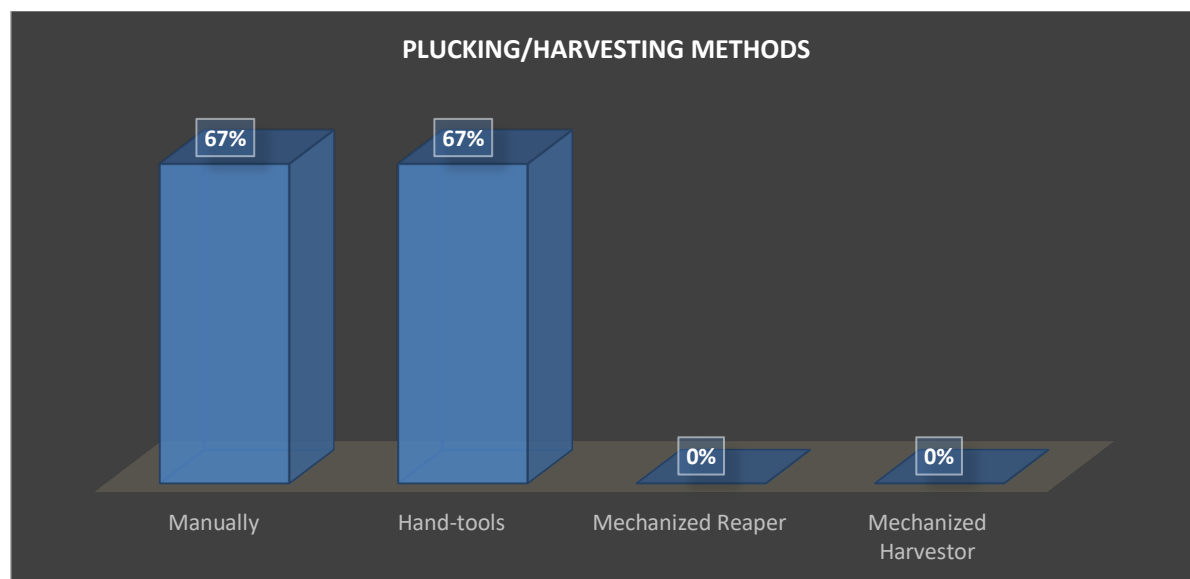
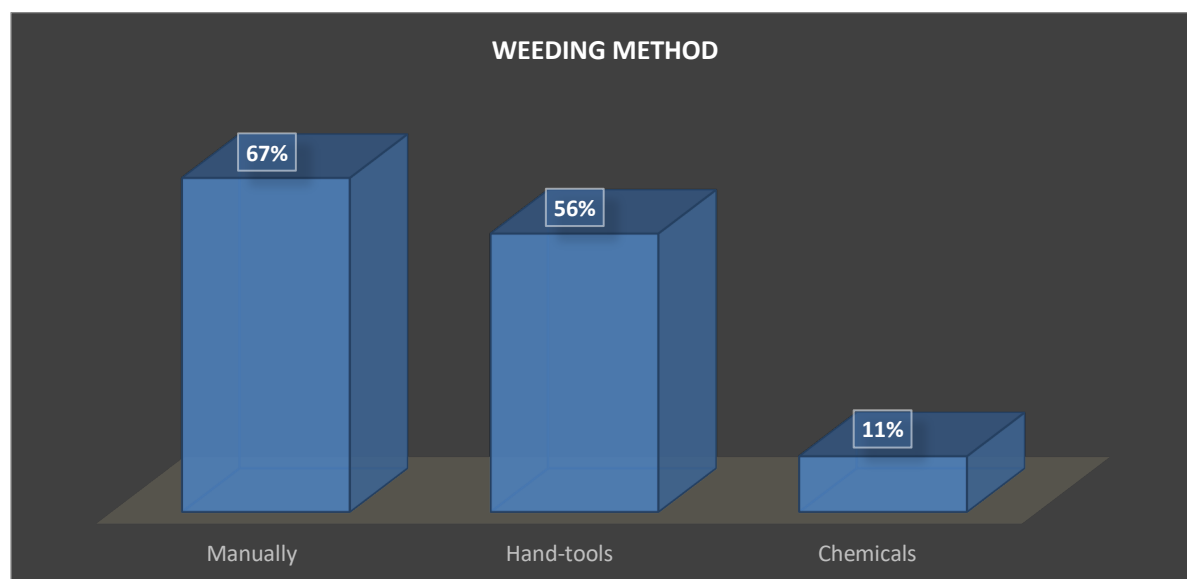


Figure 16: Weeding method

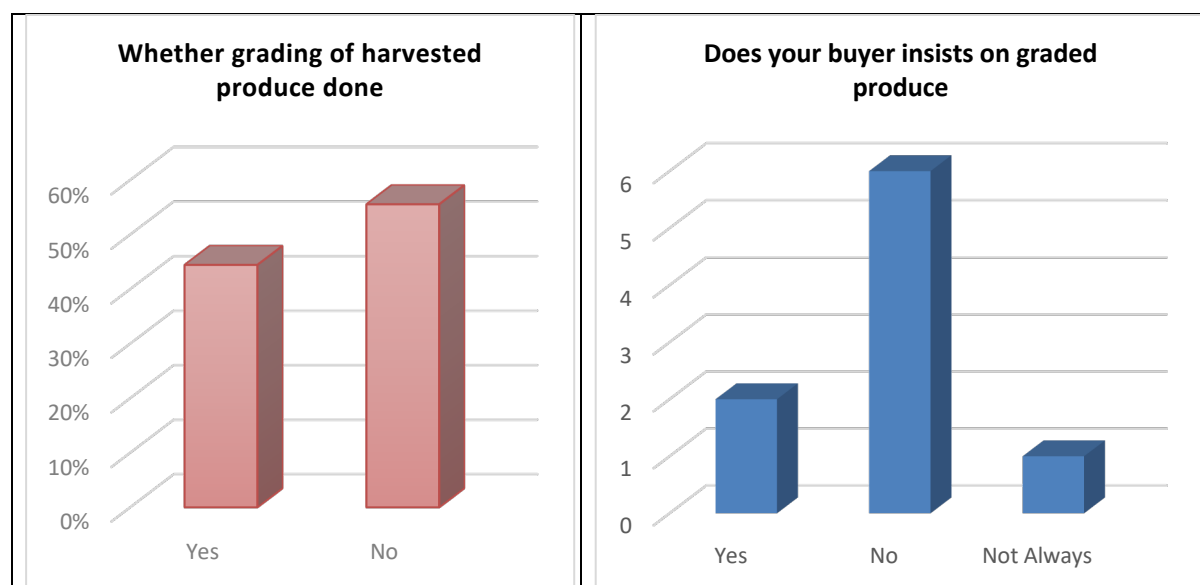


From the above two figures, it is apparent that most of the harvesting activities are still very manual or at the most use of hand tools. The reason that can be attributed is the scale of farming, as seen from the areas of cultivation, the farmers are mostly small land-holding. Investment to mechanization or increased investments will not provide the marginal utility through the increased production and sales.

4.2.3 Post Harvesting

- *Post-harvest management (Cleaning, sorting, grading, packaging etc.)*

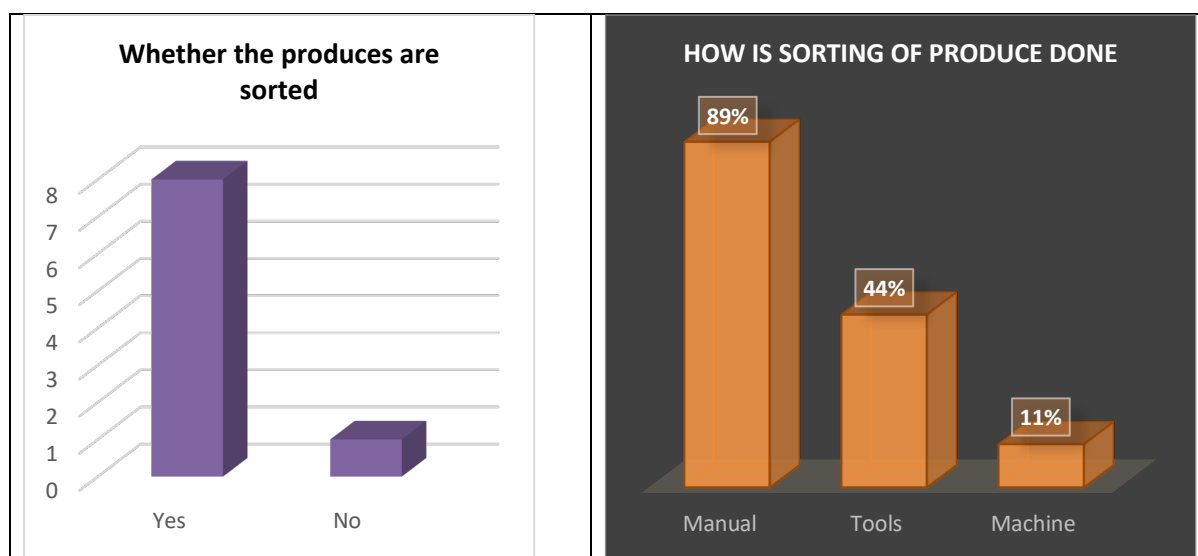
Figure 17: Prevalence of grading amongst the farmers



From the data represented in the above figures, it can be inferred that the concept of grading of the produce is not new for the farmers as well as traders. Maybe, those fruits and vegetables going to the local market where size, shape, colour etc. is not very price sensitive, grading concept is not enforced.

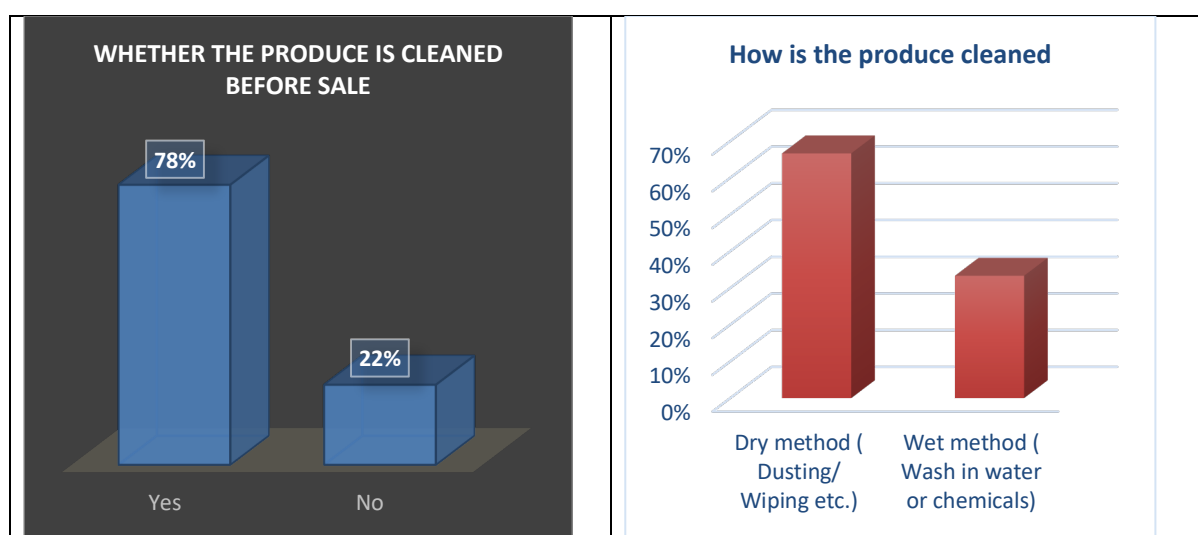
While, produce for export market or sold through retailers in price sensitive markets may see the adherence to quality norms.

Figure 18: : Prevalence of sorting amongst the farmers



Analysing the above figures, it is apparent that though grading of the produce may not be wide-spread but the sorting activity of fruits and vegetables is a common norm for the farmers. It can be inferred here that knowledge of graded produce may be limited but basic differentiator for the produce is present amongst the market forces, which leads to sorting the produce into defined categories. Though sorting is done by almost all producers but it is still highly labour intensive with minimal mechanization.

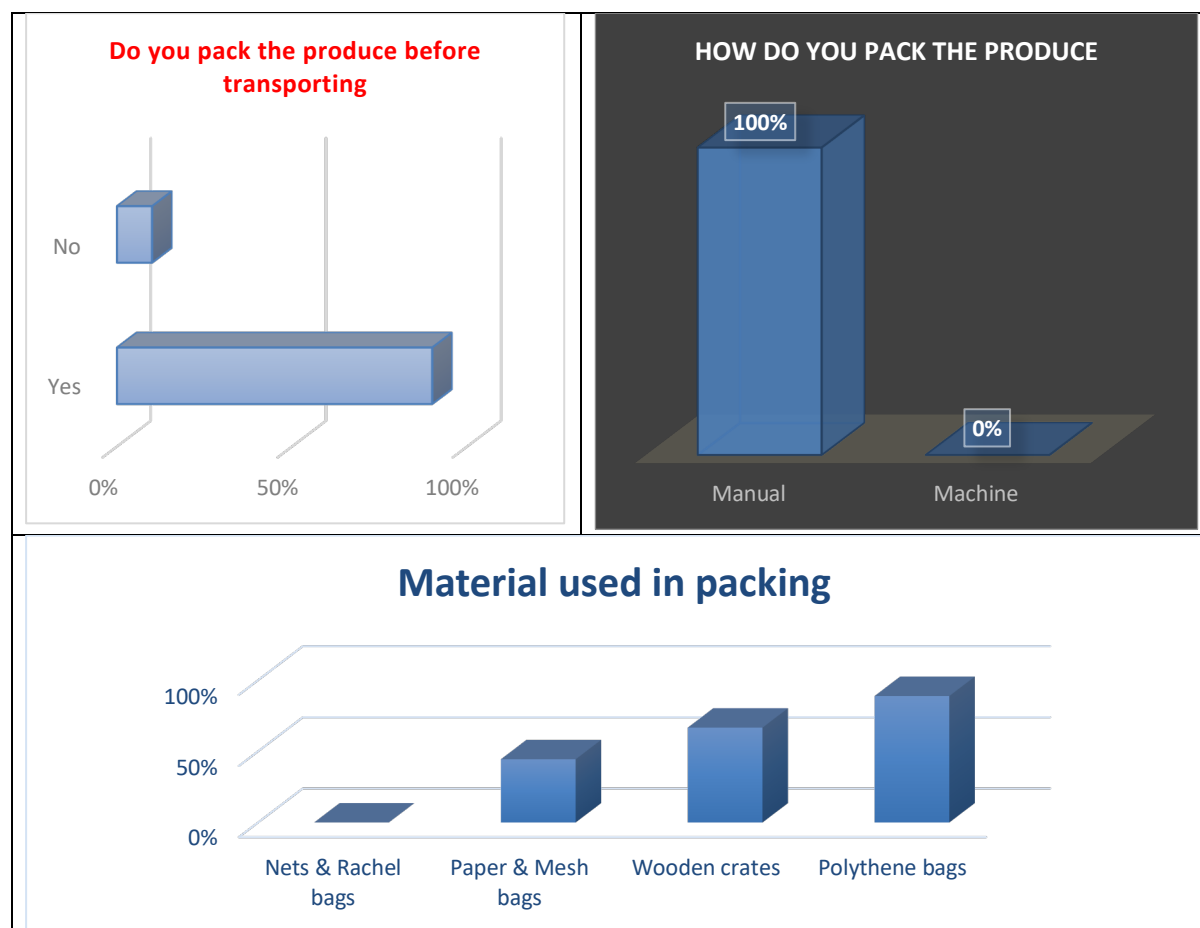
Figure 19: Prevalence of Cleaning practices of produce



As seen from the above figures, the cleaning practices after harvesting and before selling is prevalent with the method of Dry (wiping, dusting etc.) being the most popular practice.

In order to understand the packaging materials used to transport the fruits and vegetables, the respondents were interviewed on their preferences. The figure below provides an insight on the packaging methods employed by the farmers.

Figure 20: Packaging methods for fruits and vegetables used



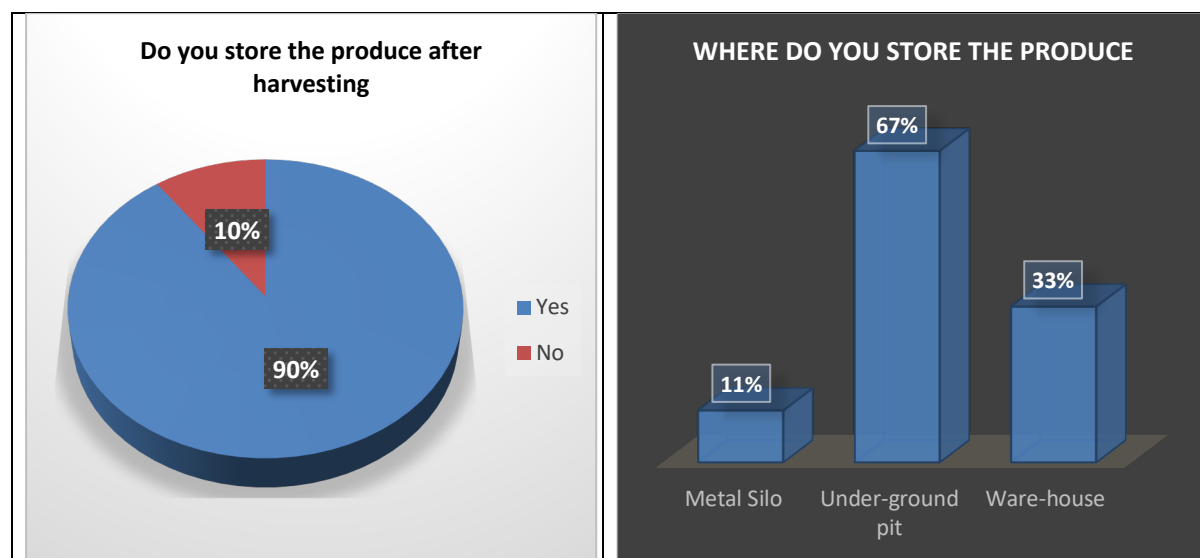
From the above data, it can be inferred that the packaging of fruits and vegetables are a mix of rudimentary and progressive. The type of packaging and material used are probably depends on the ultimate customer. Fruits and, vegetables which are soft such as tomatoes needs to be transported in wooden crates which seems to be practiced while use of plastic bags which are detrimental for vegetables since it is airtight and with temperature it is very harmful for the shelf life of the vegetables is also common. Hence, it is clear that farmers practice packing methodology based on end use as well as the increased marginal utility from use of better packaging materials.

- **Storage**

Fruits & Vegetable are crops which require special attention in harvesting and the time of sale since the shelf life is generally very short. The fruits & vegetables when they are at the right stage for plucking/harvesting, the farmer cannot delay it since it may ripen or similarly cannot be harvested earlier since it will be still under-developed for consumption or in other words the produce is unfit or unsuitable for market if harvested early or late unless the appropriate variety of this kind has been sown. Another issue that comes up while harvesting is that all the produces are not ready at the same time for harvesting and it is a period within which it needs to be harvested even though they have been planted at the same time. This causes small and marginal farmer to have small volume to trade,

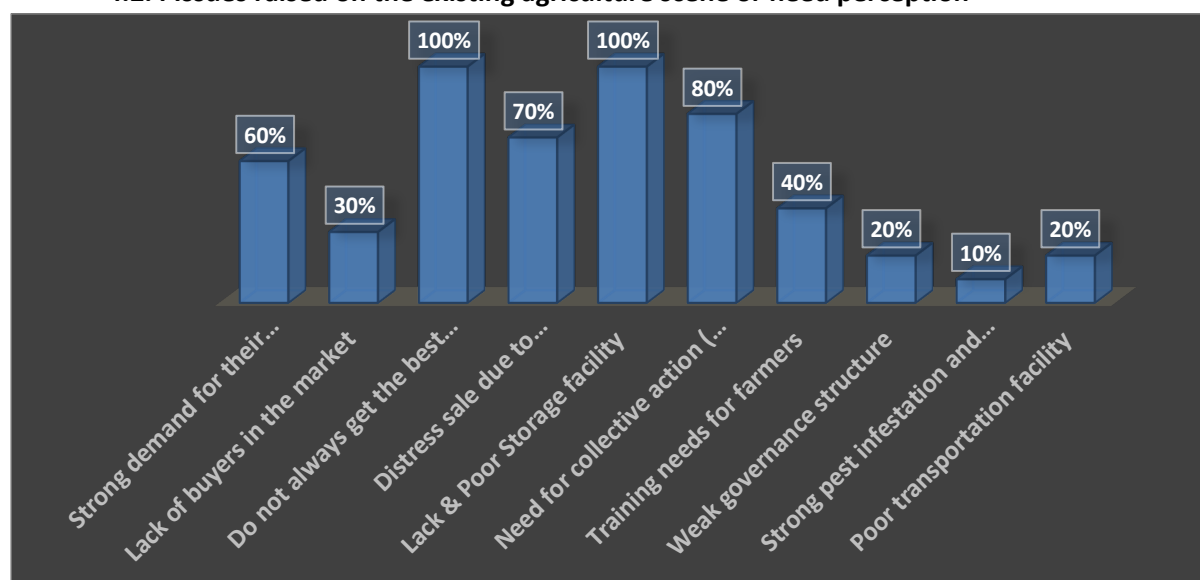
and hence it becomes costlier to carry it to a distant market for a better price. In this situation, the farmers practice some form of storage to have a larger tradeable volume or till transportation is available.

Figure 21: Prevalence of Storage practices



The above figures indicate that storage after harvesting is widely practiced and the most common method is the traditional under-ground pits. This is an area to intervene since major reason for harvest loss is due to use of traditional pits.

4.2.4 Issues raised on the existing agriculture scene or need perception



The need and issue perception of the farmers were evaluated which infers interesting results. It is very clear that cultivation of fruits and vegetables has a ready market with only 30% of the respondents saying the contrary. But interestingly, the prices/income from the production is not satisfactory with all the respondents quoting that there are numerous instances when they are not getting the best

price. 70% of the respondents has quoted that they had to resort to distress sale due to perishability nature of the produce, which in-fact validates that lack of and poor storage facility is the cause for this distress sale.

3.3 Support Services

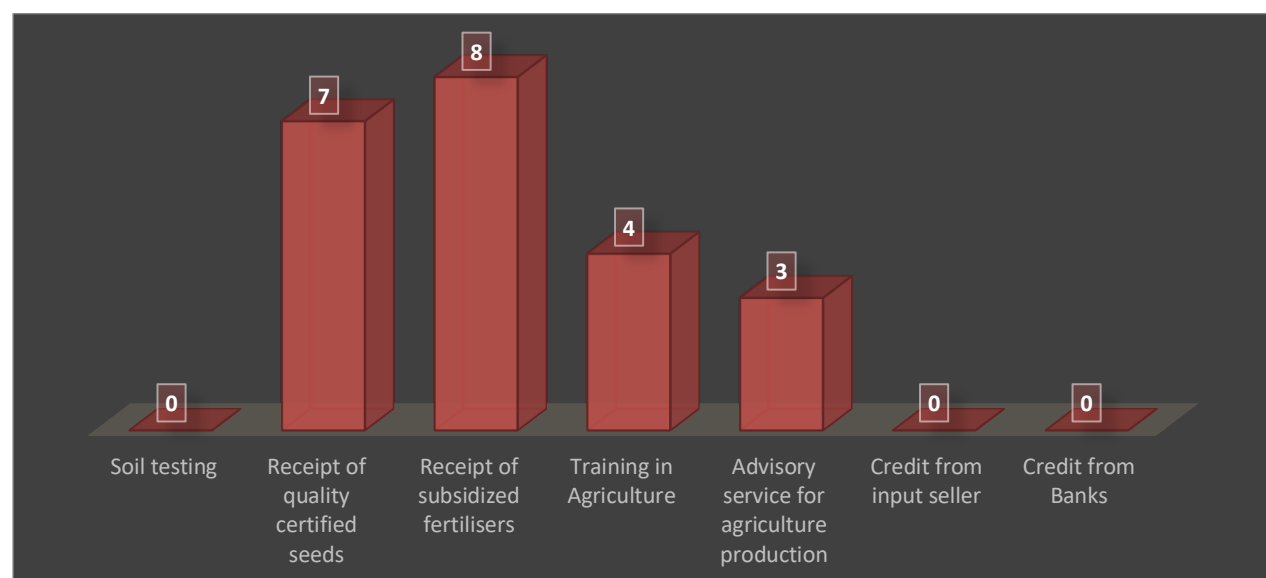
Somalia has seen many humanitarian organization playing significant role after the end of the civil war to support the evolving government structures, initially in matters of food security but gradually rebuilding the economy. Many such NGOs and Donor agencies have worked in Somalia to strengthen the Agriculture sector and some of the agencies are listed below identified through secondary research.

Table 6: List of support agencies in Somalia in the sector of Agriculture

Name of the Institution	Type of Institutions (Govt./Pvt)	Type of Services provided
Sustainable Development and Peace Building Initiatives (SYPD)	National NGO	Provision of Agricultural Inputs e.g seeds, vegetable and fruits
Somali Stability Fund (SSF)	International NGO	Provision of agricultural inputs in partnership with SYPD
Food and Agriculture Organization (FAO)	UN	- Support agricultural activities through Sustainable and Peace Building Initiatives (SYPD). - Support seed distribution in partnership with SYPD.
Danish Refugee Council (DRC)	International NGO	Provision of Agricultural inputs in order to build the capacity of local farmers both rural and semi-rural village
Norwegian Refugee Council (NRC)	International NGO	Support cultivators on Agriculture in rural and semi-rural villages in Hirshabelle, Southwest, Jubaland state
Save the Children (SCI)	International NGO	Support Agriculture activities
Mercy Corps	International NGO	Support capacity building for Somalia local farmers on Good and Agricultural Practice
Somali Agricultural Technical Group (SATG)		- Agriculture research - Technology up-gradation - Capacity building of farmers
Social-life and Agricultural Development Organization (SADO)	National NGO	Capacity Building of farmers
USAID	Donor Agency	Training of farmers and demonstration plots.
COOPI	International NGO	Distribute seeds, agricultural tools and irrigation fuel vouchers, and train farmers

The farmers were interviewed on the support that they have received from various agencies in Agriculture, the following figure captures the snapshot.

Figure 22: Support services received by farmer respondents



It can be seen from the above figure that the most common support services received is distribution of seeds and fertilizers while capacity building inputs and business related support is still absent.

3.4 Processing or productive activities

The processing activities of fruits and vegetables in Somalia is very limited with few exceptions of sesame seed oil and dry lemon. These two products have markets outside Somalia with export destinations in Middle East and Asia.

Processing of fruits for Juice is more of a micro business activity, where road-side vendors across the major towns have juice blenders and mixers to extract juice to sell in glasses. A significant amount of Juice concentrate or ready to drink fruit juice is imported to the country. Though the secondary data review has shown that few large Juice producing plants have been set-up, one in Mogadishu and two more in Somali-land.

Regarding Sesame seed processing; primarily most of the producers sell the sesame seed itself which is mostly exported. The process includes cleaning of seeds, winnowed and sieved, and dried to a moisture content of about 6-8%. Threshing is carried out to remove the seeds from their pods. Winnowing is later done to remove unwanted material from the seeds by wind. The seeds are then packed in 25 kg or 50 kg bags.

Figure 23: Harvested Sesame crops and Threshing of Sesame stalks



Source: CEFA

About 20% of the sesame harvest is used for extraction of sesame oil using simple oil expellers. The market for sesame oil is partly domestic and the rest is for exports.

It is estimated that more than 300 small and micro companies are involved in Sesame seed packing or oil expelling in Somalia.

The other product that is result of processing is dry lemon. It is primarily exported and its demand is increasing rapidly. The processing is very rudimentary without much mechanization. Dry lemon is categorised as black and yellow depending on the process used for drying. The process of harvesting and that of drying under the sun are the same for both black and that of yellow dry lemon. But the difference comes when after fully dried in the sun, in-case of black dry lemon, a very hot water steam is poured on the lemon which turns it into black.

Figure 24: Images of yellow and black dry lemon



3.5 Trading Channel & Sales

Primarily the trading channel operative in Somalia in the Fruits and Vegetable sub-sector can be classified according to the following channels of trading.

A)



B)

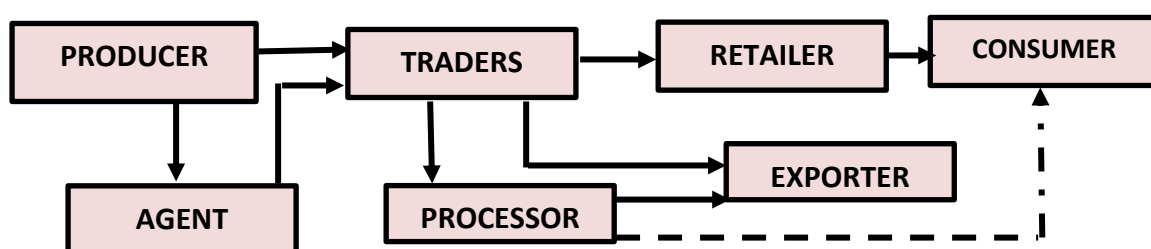


Figure 25: Use of trading channel by producers

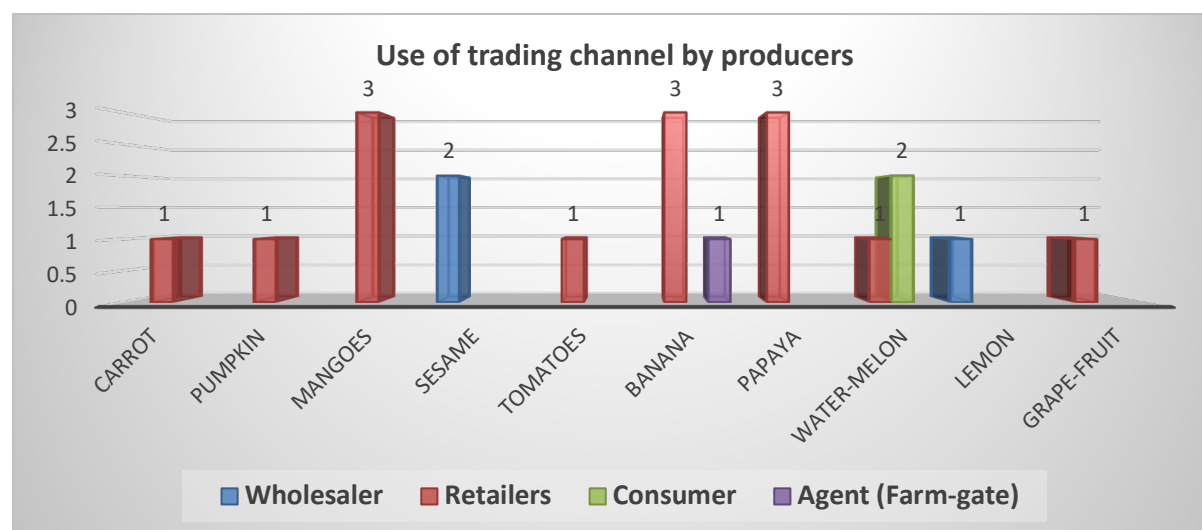
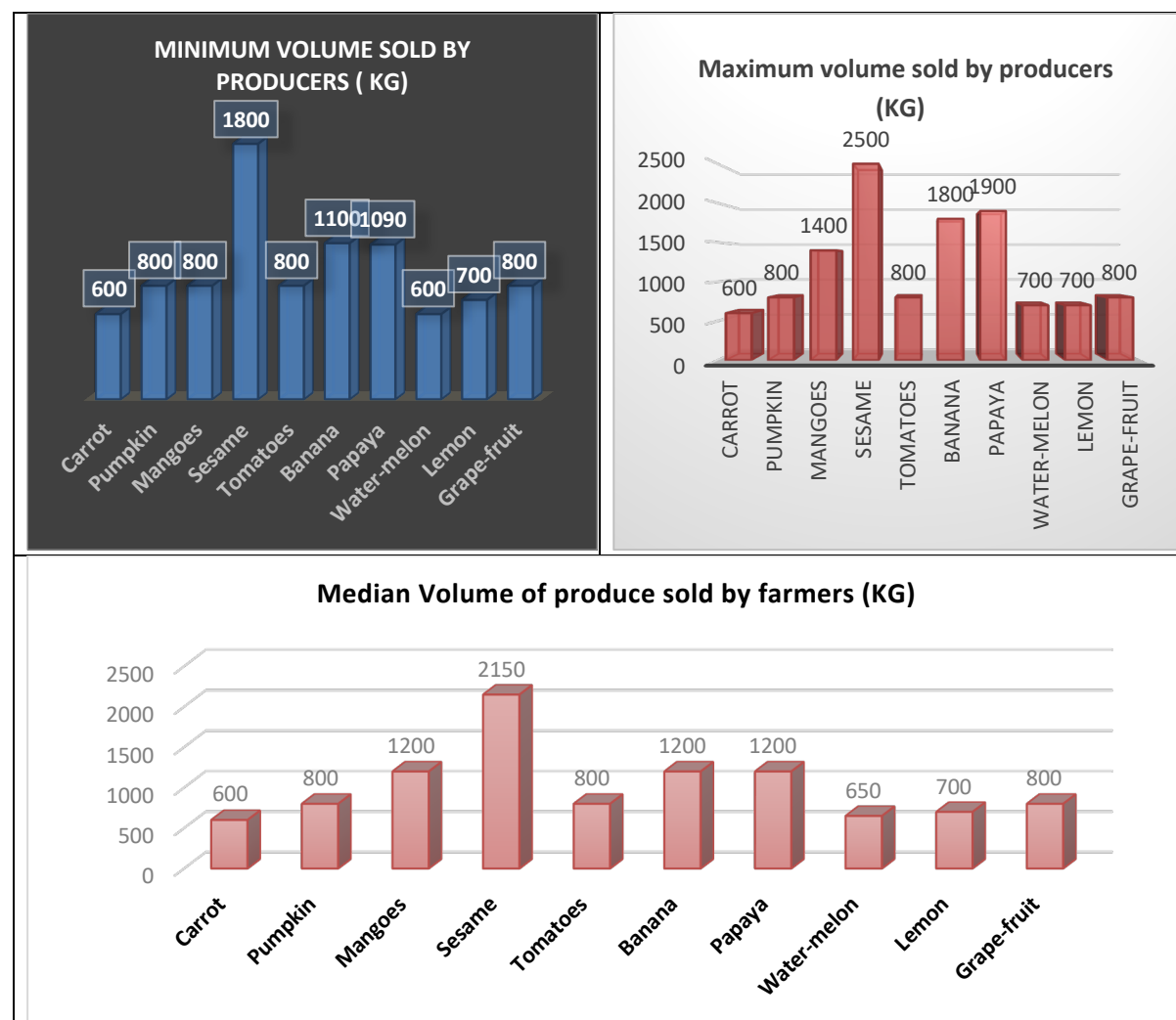
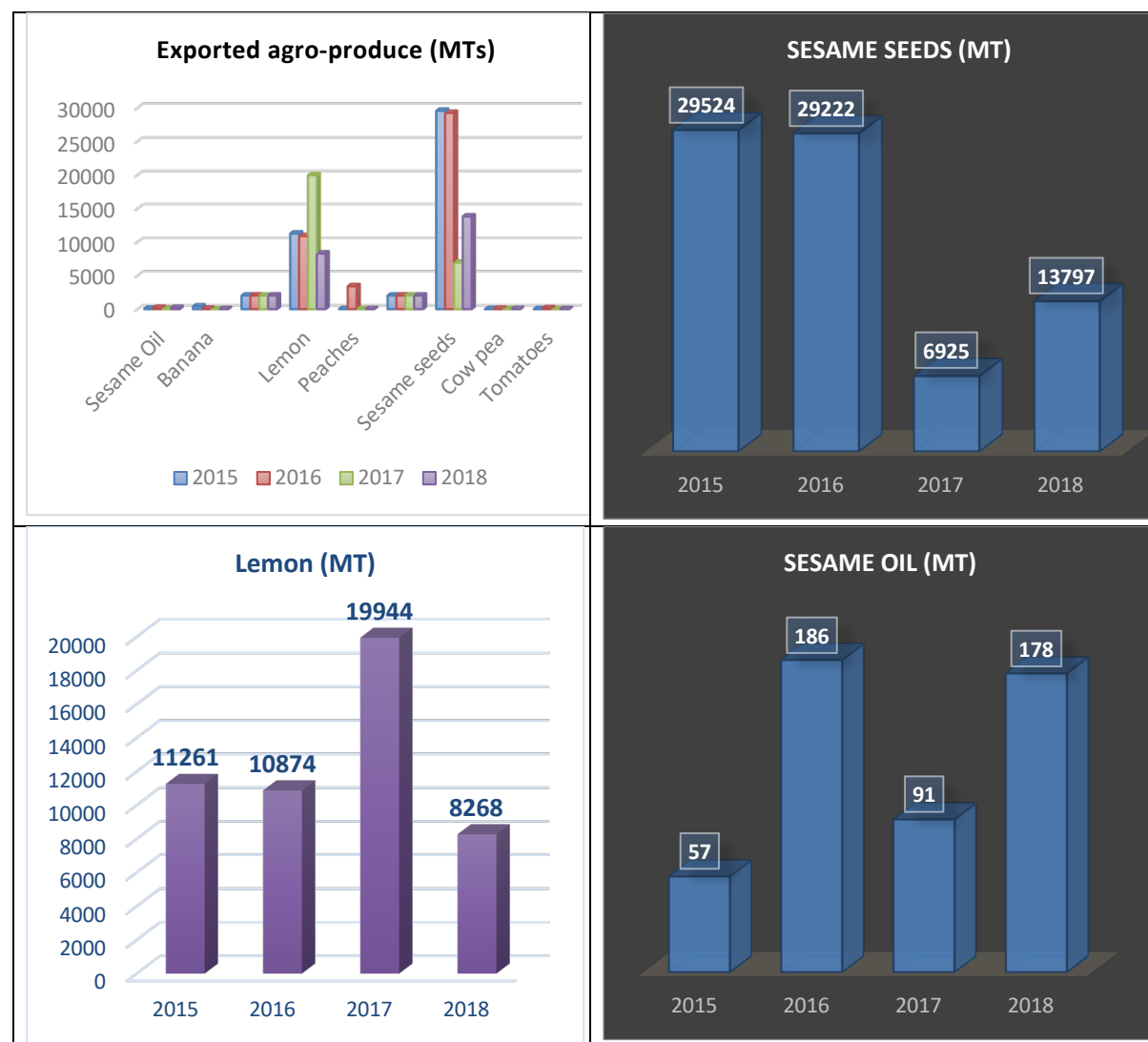


Figure 26: Volume of produce sold



While analysing the above figures, it is apparent that most of the selling channel for the farmers are the retailers (or local traders) since most of the produce is for the local consumption or at the most goes to the regional markets. While the produces which are sold through wholesalers (**Sesame and Lime**) is primarily for the export markets. The statistical reference to exports is being provided in the next figure. The wholesalers seem to be aggregator for the exporters for these produce.

Figure 27: Exported Agro-produce from Somalia



It can be seen from the figures that lemon, sesame seed and sesame oil are the major export items, while banana which was a prized item of exports in pre-civil war era completely went out of the picture. Many of the reasons for this decline, are destruction of banana farms during the civil war, disruption of the trade channels, strict phyto-sanitary restriction in Europe etc.

Based on the available price data of the sesame seed oil, which is being presented below, a special analysis has been done to understand the market movement.

Figure 28: Price of local sesame oil

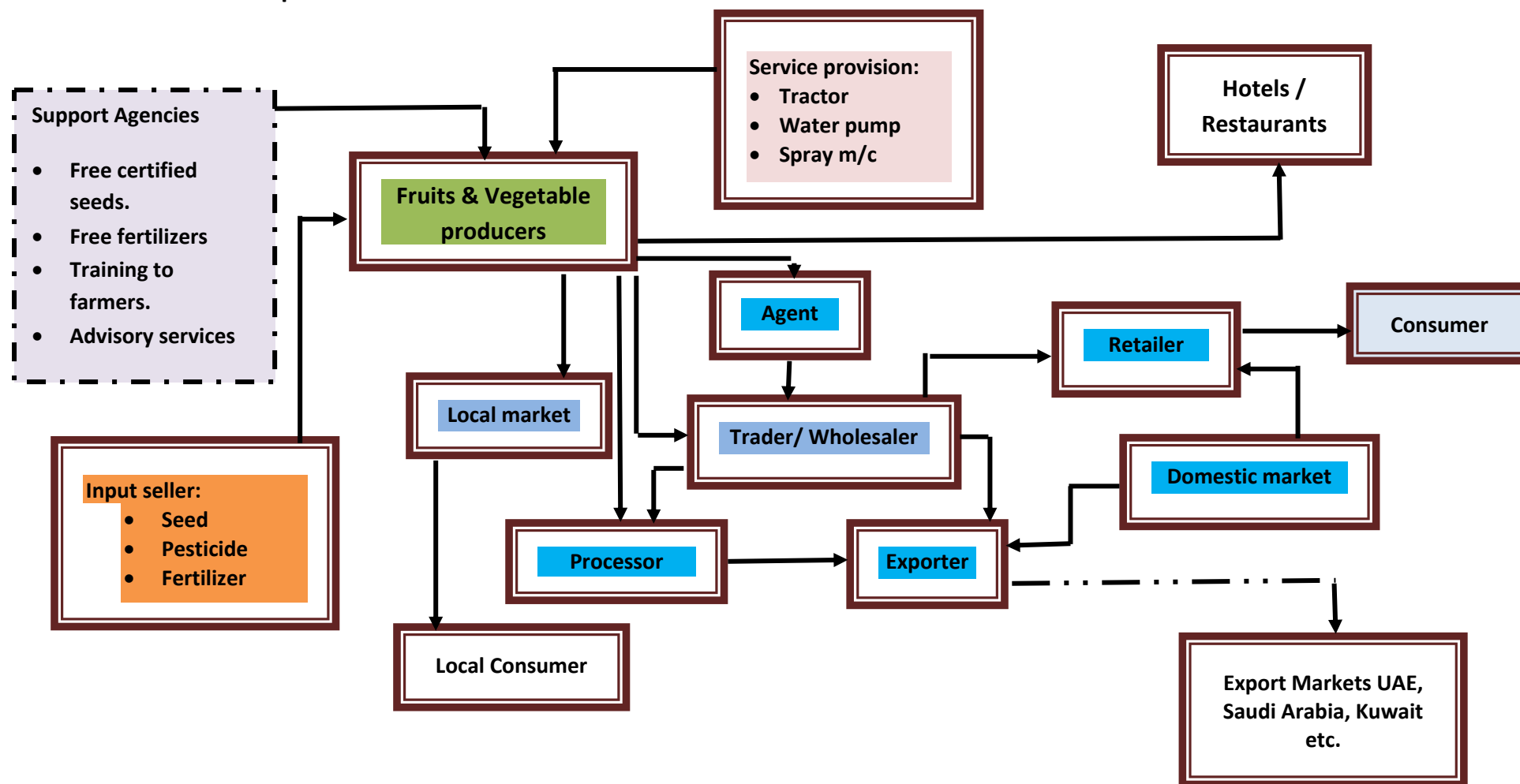


Source- FSNAU

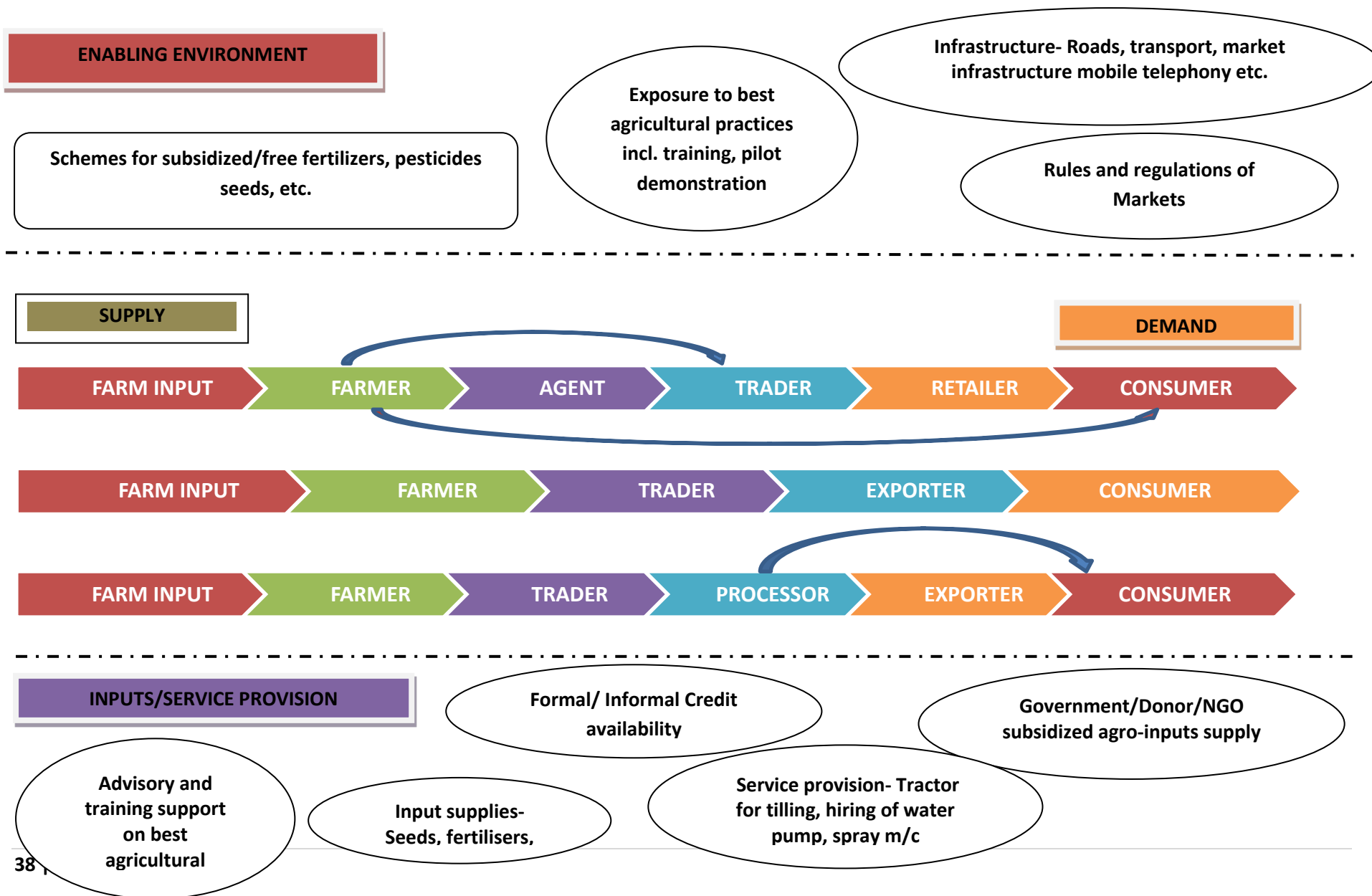
It can be seen from the above figures that the unit price of local sesame oil is gradually increasing in the studies regions with time. The unit rate in 2017 is lower compared to 2018 and later to 2019 across the sampled regions. It can be inferred two things from the findings, either exports of sesame oil or seed is increasing or the domestic consumption of sesame oil is increasing in the country, irrespective of seasonal fluctuation of price.



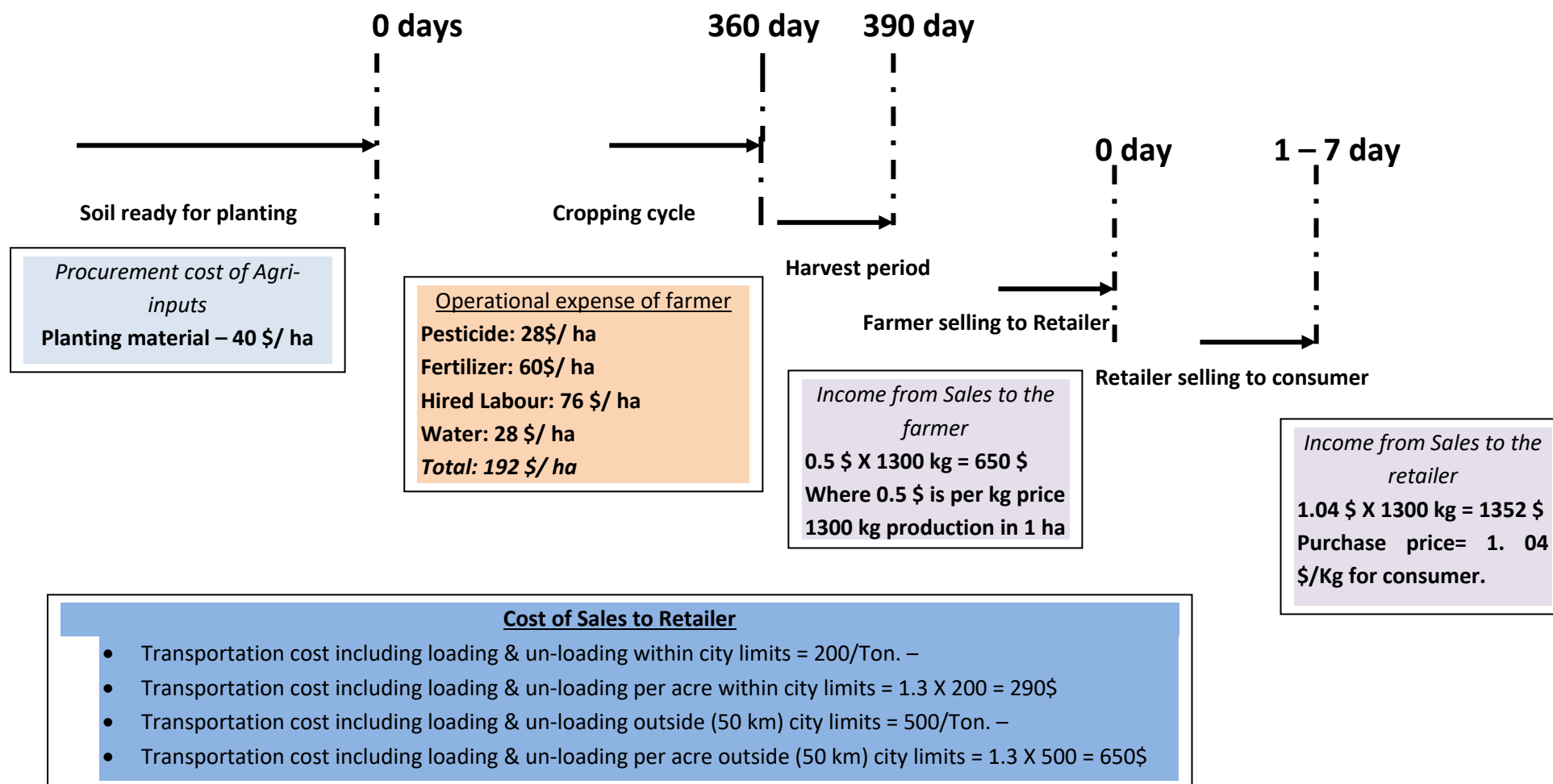
3.6 Sub-sector map



4. VALUE CHAIN FRAMEWORK FOR FRUITS & VEGETABLES SUB-SECTOR

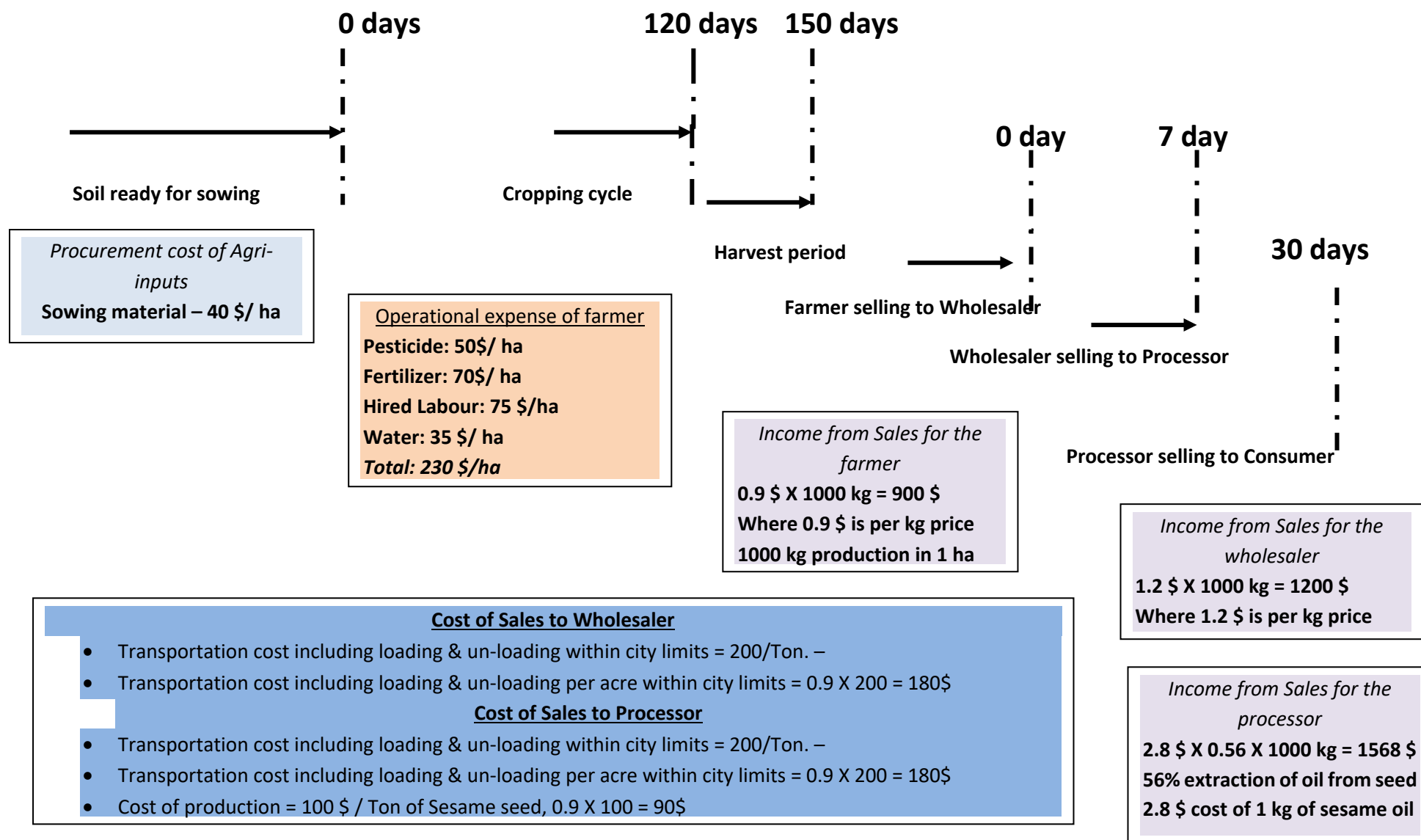


Value addition in Banana value chain (Production and sale from 1 hectare)



Producer			Retail seller		
- The farmer prepares soil for sowing the seed or planting the seedling. - The agri-inputs for farming is mostly from local retailers. - Most of the soil preparation is done through hired tractors.	Depending on the crop cycle, which is approximately 390 days for bananas, the farmer works on fertilization, disease prevention, weeding, irrigation using either hired labour or own labour.	- The harvesting period will be 30-45 days after first crop is harvested. - The farmer transports the harvested produce to the market to sell directly or through retailers collecting at the farm gate. - The harvested produce has to be sold within the shelf life period or the price will fall.	- Retailer purchases the produce either from rural primary market or at the farm gate of the farmer. - Retailer has to dispose the purchased bananas as soon as possible for the fear of getting spoiled.	- Retailer transports the produce to the place where it will undertake retail sale. - The retailer sells directly or does sub-retailing of the produce.	- The retailer will be selling bananas at price significantly high then at what he buys. - The volume per customer is small.
Gross value received on sales		650 \$	Gross value received on sales		1352 \$
Cost of the production		40 \$ + 192 \$ = 232 \$	Cost of Procurement		650 \$
Cost of Sales		-	Cost of Sales		290 \$
Value added		418 \$	Value added		412 \$
Time period		365 to 390 days	Time period		1 to 7 day

Value addition in Sesame seed value chain (Production and sale from 1 Hectare)



Sesame Producer			Wholesaler			Oil Processor		
PROCURE	GROW	SALE	PURCHASE	DISTRIBUTE	SALE	PURCHASE	PROCESS	SALE
- The farmer prepares soil for sowing the seed. - The agri-inputs for farming is mostly from local retailers. - Most of the soil preparation is done through hired tractors.	Depending on the crop cycle, which is approximately 120 days for tomato, the farmer works on fertilization, disease prevention, weeding, irrigation using either hired labour or own labour.	- The harvesting period will be 30 days after first crop is harvested. - The farmer sells the harvested produce to the wholesalers.	Wholesaler purchases the produce either from rural primary market or the farmers directly.	Wholesaler transports the produce to the processors.	Wholesaler will be selling sesame at price significantly high then at what he buys.	Processor either buys the sesame seeds from the primary market or supplied at the doorstep by the wholesaler	Sesame seeds are grinded to expel the oil content using grinding machines. Extraction percentage is 56%.	Sesame seed oil is then supplied to retail outlets and sold to consumer.
Gross value received on sales		900 \$	Gross value received on sales		1200 \$	Gross value received on sales		1568 \$
Cost of the production		40 +230 = 270 \$	Cost of Procurement		900 \$	Cost of Procurement		1200 \$
Cost of Sales		-	Cost of Sales		180 \$	Cost of Sales		180 +90 = 270\$
Value added		630 \$	Value added		220 \$	Value added		98 \$
Time period (days)		120 -150	Time period (Days)		7	Time period (Days)		15-30

5.1 Governance

Value chains involve regular interactions between the various actors implying repetitiveness in their relationships. Governance ensures that interactions between firms along a value chain exhibit some reflection of order rather than being simply random. Value chains are governed when parameters requiring product, process, and logistic qualification are set which have consequences up or down the value chain encircling package of activities, actors, roles, and functions.

Governance can be categorised into the following types; **legislature** (making the rules), the **executive** (implementing the rules) and the **judiciary** (monitoring the conformance to rules). This governance can be exercised by parties internal to chain or external to chain.

As a part of the value chain analysis for the fruits & vegetable sub-sector in Somalia, the chain governance was looked into;

	Exercised by parties internal to chain	Exercised by parties external to chain	Implications
Legislative governance	- Quality standards of the vegetables (size, colour, state of ripeness etc) from the farmers are governed by the traders who purchase from them. Strong - The price fixations of the produce in the markets are determined based on demand – supply situation in the market on the day of transaction. Strong	- Market committee/ Check points fixes the taxes & commissions to be collected for the transactions taking place and goods movement. Strong - Local authorities'/ Agriculture department have formulated rules for retail suppliers of seed, fertilizers, pesticides etc to ensure quality products. Weak	- Farmers have to be particular about the quality of the produce taken to the market. - The price of the produce to be received by the farmer depends not only on the quality but also on the demand and supply situation which is beyond the farmer's control.
Executive governance	- Commission and taxes fixed by the market authority / Check points are collected by the commission agents from the traders and the farmers. Medium - The agreed price of the vegetables is paid by the buyers to the farmer directly. Strong	The local agriculture/ horticulture department/ NGOs/ humanitarian agencies are making seeds/ fertilizers available to the farmers at subsidized rates or free. Strong	- The trade transactions are clearly defined by the market forces and the possibility of the farmer getting cheated is minimal. - The benefits designed to support poor farmers are received but the outreach of the support needs to be analysed.
Judicial governance	The quality/ produce conformation is checked by the traders while procuring them from farmers/ local traders. Strong	Agriculture department monitors quality conformation by the input suppliers. Weak	Lax governance on agri-input quality adherence exposes the farmers to ineffective agri-inputs harming them in production.

5.2 Inter-firm cooperation

Inter firm cooperation is defined as joint activities undertaken by the various stake-holders in the value chain. Cooperation can be subdivided in *horizontal linkages*, i.e. cooperation among similar types of enterprises, for example vegetable producers, and *vertical linkages*, cooperation between enterprises at different positions in the value chains, for example the local trader and vegetable.

Horizontal linkages refer thus in particular to the regular contacts and relationships between peer enterprises within a specified area, more specifically, it refers to joint activities of such firms, i.e. joint purchases, joint usage of equipment, joint contracts and joint learning

On the one hand, vertical linkages refer thus to contracting and subcontracting relationships between enterprises of different types in terms of scale.

5.2.1 Horizontal linkages

Horizontal linkages, or the cooperation among similar types of stake-holders, are weak. In case of fruits and vegetable producers, recognized horizontal linkages are very minimal almost to the point of nil. Some cases of inter-firm cooperation amongst fruits and vegetable producer are observed in terms of sharing transport mode to carry the produce to the market. Some other linkages though not at cooperation level but more for service provision is noticed, the owners of tractors frequently rent it out to other farmers or rent out spraying machine.

At the traders' level too, horizontal linkages are not apparent. The retailers or wholesalers buying from the farmers in the market do not have any established inter-firm cooperation.

It has been reported that some associations at the regional level for producers and traders have been formed. Concrete steps of cooperation was not reported during the study.

5.2.2 Vertical linkages

In general terms, stake-holders in the value chain have rather weak vertical linkages. The major reason behind it is the state of the sub-sector as such. It is much localised and mostly un-organized. The need and advantage of building vertical linkages are still evolving and is in the dormant stage.

The most common form of vertical linkages observed is receiving advance amount by the farmers from the traders in procuring the produce.

5.3 Structure of labour market

The labour requirement in the fruits & vegetable cultivation is very high, starting from land preparation to harvesting, with intermittent inter-culture. The present usage pattern of labour is mix of self or family and hired labourers. This is possible due to small land utilization for fruits and vegetable cultivation. Primary data from respondents have indicated that external labour is used at times mostly for land preparation, inter-culture and harvesting with the requirement varies from specific to few types of fruits and vegetables. This is more apparent in case of vegetables than fruits. The labour force used here is mostly local from the village or surroundings.

The markets have un-regulated established labour force, whose work includes loading and unloading of fruits & vegetable bags, weighing of the vegetables etc. There are standard rates for these functions, which needs to be paid by the user of the service. The availability of this labour force is from areas near to the markets. This labour force is mostly independent.

6 FINDINGS & KEY ISSUES

Most of the farmers involved in Fruits and Vegetables sub-sector except few commercial farms are marginal farmers. The marginal utility of increasing production through use of mechanised farming and quality agri-inputs is not very remunerative. The production has to have aggregation for economy of scale where these inputs and mechanization can be fruitful.

The existing value chain of the fruits and vegetable sub-sector is highly under-developed and lots of gaps within. One of the major drawback for the farmers/ traders is the lack of developed storage facilities to increase the shelf life of the produce. This leads to distress sale and the farmers do not get good prices. In addition, lack of processing facilities barring few produce like lemon & sesame seed makes the farmers go for limited production.

Since most of the produce are for the local market, much attention is not given towards skill-sets on grading/ sorting, handling during harvesting, packing, transporting, stacking in the trucks etc. due to which these skill-sets of the workers have not been developed. If more processing facilities and farms exporting are established, the existing skilled work-force will not have the capacity to provide efficient service which will result losses due to produce damage.

It has been reported that more than 300 micro enterprises are in Banadir region itself, which uses grinding stone to extract oil from sesame seed. Somalia is one of the few countries in Africa, which grows sesame seed and exports mostly seeds and small amount of sesame oil. There is scope of using better techniques such as mini oil expellers to extract oil from sesame seed which will increase the extraction percentage. Capacity building will be required in the existing skill force or create vocational trained youths on the skill sets of operating mini expellers, on how to grade and clean sesame seeds and also packaging.

The knowledge of packaging material and best practices of packaging, as well as the skill sets of under-taking modern techniques of packaging is missing in the sub-sector. This is owing to the fact that most of the agro-produce are primarily focussed towards local domestic market and investment in packaging for the farmer or the trader is an inconsequential requirement. Hence, these skill-sets if not developed now, will create a vacuum later when enterprises and investment flows in the area of exports and processing. There will be lack of trained skilled work-force who could contribute in the packaging part, thereby, reducing damages, and improving the aesthetics of packaging, increasing the shelf life.

Since, many of the fruits and vegetables do not have much potential for exports or high margin if sold as whole fruit/vegetable, there are possibilities of extraction of Juice from some of these fruits and vegetables. Though, except for one or two modern units of Juice extraction have been established, it could be area of possible investments coming in next couple of years in view of large quantity of Juice is imported to the country, thereby promising scope for local content. But, as of now, Somalia do not have skilled work-force who can handle operations of mechanized Juice extraction units.

7 RECOMMENDATION ON AREAS for SKILL SET UPGRADATION

A) POST – HARVEST

Grading, Sorting and Packaging	
Areas of Skilling	Remarks
Norms of grading of fruits and vegetables	These training modules will be targeted at the workers working on farms producing fruits and vegetables for export markets.
How to do sorting based on grades	
Knowledge on operation of sorting machines – particularly for tomatoes, potatoes	
Techniques for dry and wet cleaning of fruits and vegetables	
Packaging, Storage and Transportation	
Areas of Skilling	Remarks
Knowledge on various packaging materials and their strengths and weaknesses.	These training modules will be designed for creating skilled workers in the field of packaging.
Technique and knowledge on packaging for domestic market requirements.	
Technique and knowledge on packaging for export market requirements. • Corrugated Fibreboard boxes • Plastic corrugated boxes • Wooden Box • Plastic crates • Sacks • Palletization	
Technique on how to store harvested fruits and vegetables to slow down deterioration of shelf life.	
Handling, stacking of fruits and vegetables in trucks, loading, unloading to minimise damage to the fruits and vegetables.	These training modules will be targeted at the workers working on farms producing fruits and vegetables for commercial purpose.

B) PROCESSING

Areas of Skilling	Remarks
Technique of using mini oil expeller for cold press of sesame seed to extract oil.	Most of the units use grinding machine which results inefficient extraction. The training modules will be geared towards vocational training inputs for preparing skilled workers who will have the knowledge to operate Mini oil expeller.
Technique for Juice extraction process in a semi mechanised unit.	Gradually, units will be coming up in Somalia, which will start mechanized Juice extraction units and bottling of Juice. This training module will be geared towards creating a skilled work-force that can be of high demand in coming years.

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